Vol. **000**, 1–23 (2023) Preprint 3 November 2026 Compiled using LaTeX style file v3.0

On the statistical theory of self-gravitating collisionless dark matter flow

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

Dark matter behaves as a self-gravitating, collisionless fluid, and its flow changes character with scale. We develop the statistical theory of its second-order velocity statistics and test it against the Virgo SCDM N-body simulation from $z = 10$ to $z = 0$, measuring every statistic directly from particle pairs rather than from a mesh. We first derive exact kinematic relations between the correlation, structure, dispersion and spectrum functions of three idealized flows (incompressible, constant-divergence and irrotational) and show that incompressible and constant-divergence flows share the same relations for even-order correlations. Applying these relations to the simulation shows that the peculiar velocity is a constant-divergence flow on small scales, where particle pairs sit in the same halo, and an irrotational flow on large scales, with the transition near $r_t \approx 1.3\,\mathrm{Mpc}/h$. Because the flow is collisionless, the longitudinal velocities of two particles at vanishing separation are only partly correlated: we estimate a limiting correlation coefficient $\rho_L = 1/2$, compared with $\rho_L = 1$ for incompressible flow. On large scales the transverse velocity correlation is exponential, $T_2 \propto e^{-r/r_2}$, with a time-independent comoving scale $r_2 \approx 23.1\,\mathrm{Mpc}/h$ that may be related to the horizon size at matter–radiation equality; all other velocity, density and potential statistics on these scales then follow analytically. On small scales the longitudinal structure function follows a one-fourth law, $S_2^l \propto r^{1/4}$, from which the remaining small-scale statistics follow. Vorticity is negatively correlated for separations between 1 and $7\,\mathrm{Mpc}/h$, and divergence for $r \gtrsim 33\,\mathrm{Mpc}/h$, which produces a negative density correlation on those scales.

Key words: dark matter – large-scale structure of Universe – methods: numerical – methods: statistical

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1 INTRODUCTION

The existence of dark matter is supported by many astronomical observations [1, 2]. In the standard ΛCDM (cold dark matter) paradigm of cosmology, the amount of dark matter is about five times that of baryonic matter [3, 4, 5]. Therefore, the flow of self-gravitating and collisionless dark matter possesses the widest presence in our universe. Cosmic flow velocity is a key component of self-gravitating collisionless fluid dynamics (SG-CFD) for dark matter. Many insights were obtained from the velocity field on structure formation and evolution and the interpretation of observational data. Due to the complicated nonlinear nature, N-body simulation is an invaluable tool to study SG-CFD, capture the very complex gravitational collapse and many effects beyond Newtonian approximations [6, 7, 8, 9]. The complete review of N-body simulation methods for large-scale structure and galaxy formation was discussed by Angulo and Hahn [10] and Vogelsberger *et al.* [11]. Recent improvements have also been presented with the information-optimized parallel algorithm [12] and the fast multiple method [13] for N-body simulations, and a new approach to generate initial conditions for N-body [14].

Traditionally, two distinct approaches have been applied to study the velocity field from the N-body system, i.e. a halo-based approach and a statistical approach. In the halo-based approach, all haloes are identified in N-body simulation with various statistics defined for particle velocity in haloes and velocity of haloes. Using this approach, velocity distributions of dark matter particles that max-

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imize system entropy can be rigorously formulated [15]. An alternative strategy without identifying haloes, i.e. a statistical approach, works with statistical measures, such as the correlation and spectrum functions, to make a statistical statement about SG-CFD. In this approach, haloes are not explicitly considered. Instead, we focus on the scale and redshift dependence of these statistical measures, which are crucial for structure formation and dynamics on both small and large scales [16, 17, 18]. However, it is not a trivial task to extract and characterize the statistics of velocity field from N-body simulations. Some fundamental problems are as follows:

(i) The velocity field is defined everywhere in the entire phase space, but is only sampled by N-body simulations at discrete locations of particles. This sampling has a poor quality at locations with low particle density that leads to large uncertainties on the measure of the velocity power spectrum [19].
(ii) The velocity field can be multi-valued and discontinuous after shell-crossing due to the collisionless nature. The traditional statistical approach involves computing the power spectrum of the velocity (and its gradients) in the Fourier space [17, 20, 21]. It uses cloud-in-cell (CIC) schemes [22] or triangular-shaped-cloud (TSC) schemes to project density/velocity fields onto the regular mesh, which unavoidably introduces a finite sampling error [23, 24].

Since both the real- and Fourier-space contain the same information, there is no reason for preference of one over the other, except for the computational cost. Directly working on the density/velocity fields in real space avoids an additional layer of systematic errors and information loss due to the mapping of particles onto structured grid and the conversion from Fourier space to real space.

The statistical theory for incompressible flow was developed for isotropic and homogeneous turbulence [25, 26, 27, 28]. However, the self-gravitating collisionless dark matter flow is intrinsically different. The cosmic peculiar velocity is not incompressible (vanishing divergence) on all scales. Instead, on small scales, virialized haloes have a vanishing (proper) radial flow and the peculiar velocity $\mathbf{u}$ satisfies a constant divergence condition or $\nabla\cdot\mathbf{u} = -3Ha$, where H is the Hubble parameter and a is a scale factor [29]. While on large scales, flow becomes irrotational or $\nabla\times\mathbf{u} = 0$. The nature of two different types of flow on small and large scales and the transition between them are the key to understand the velocity field in dark matter flow. Therefore, it is necessary to develop a complete set of kinematic relations between various statistical measures for different types of flow.

Physical picture. The two regimes have a simple origin. Inside a virialized halo the proper (physical) size no longer changes, so in comoving coordinates every halo shrinks at the rate set by the Hubble expansion; the peculiar velocity of its particles therefore converges uniformly, $\nabla\cdot\mathbf{u} = -3Ha$, independent of position inside the halo. Two particles closer than a typical halo size usually share a halo, so small-separation statistics measure this constant-divergence flow. On scales much larger than haloes, matter moves coherently in response to linear density perturbations; the peculiar velocity is then the gradient of the gravitational potential and carries no vorticity (an irrotational, potential flow). The kinematic relations of Section 3 turn these two statements into testable relations between measured functions, and Section 4 shows that the simulation switches from one to the other near $r_t \approx 1.3\,\mathrm{Mpc}/h$.

This paper focuses on the two-point second order statistical measures of velocity field for SG-CFD. Using second order velocity statistics fully extracted from N-body simulation without projecting onto structured grid, we are able to:

(i) demonstrate that dark matter flow is irrotational on large scales and of constant divergence on small scales (Fig. 6);
(ii) identify an exponential form of the transverse velocity correlation on large scales (Eq. (110)), from which all other statistical measures of the density, velocity and potential fields can be derived analytically;
(iii) identify a one-fourth scaling law for the second-order longitudinal structure function on small scales ($S_2^l(r) \propto r^{1/4}$, Eq. (137)), from which all other small-scale measures follow;
(iv) estimate a limiting longitudinal velocity correlation coefficient of 1/2 on the smallest scale ($r \to 0$; Section 3.3.1 and Fig. 17).

This paper is organized as follows: Section 2 introduces the N-body simulation data used for this paper. Section 3 introduces the two-point second order statistical measures and develops the fundamental kinematic relations between different statistical measures. Generalization to high and arbitrary order is presented in a separate paper [29]. Section 4 presents the statistical measures from N-body simulations, followed by the corresponding models on small and large scales in Section 5. Section 6 summarizes the results. Table N lists the principal symbols.

2 N-BODY SIMULATIONS AND NUMERICAL DATA

The numerical data are publicly available and generated from the N-body simulations carried out by the Virgo consortium. A comprehensive description of the data can be found in [30, 31]. As a first step, the current study was carried out using simulation runs with Ω=1 and the standard CDM power spectrum (SCDM) that started from $z = 50$ to focus on the matter-dominant self-gravitating flow of collisionless dark matter. Similar analysis can be extended to other simulation with different cosmological model assumptions and parameters in the future. The same set of data has been widely used in studies from clustering statistics [31] to formation of cluster haloes in a large-scale environment [32], and in the testing of models for halo abundances and mass functions [33]. Some key numerical parameters of the N-body simulation are listed in Table 1. Figure 1 presents several simulation snapshots at different redshifts showing the dynamics of self-gravitating dark matter leading to the formation and evolution of structures.

To analyze the nature of flow of dark matter only, we first focus on the cosmological model with dark matter only and exclude any other effects. This situation is more relevant to the early matter-dominant universe. This will also establish a benchmark for future testing and comparison. The simulation campaign from the Virgo Consortium also includes data for different cosmological models (LCDM, OCDM, and tCDM, etc.)[30]. By gradually increasing the complexity, the approach described in this paper can be applied similarly to identify effects of various cosmological parameters on the dynamics of dark matter flow.

In addition, in this paper, we propose an approach to directly compute the real-space statistics by pairwise averaging over all particle pairs on a given scale. This approach can maximally employ the information contained in N-body simulation and generate complete statistics on all scales without involving any projection kernels (e.g. CIC). On the other hand, it is also computationally intensive to identify all particle pairs on all scales. The selected N-body simulation has a relatively low resolution when compared with recent cosmological simulations. However, this enables a computationally affordable direct extraction of real-space two-point statistics. With increasing computing power, the same approach can be similarly applied to other simulations with higher resolution. The pair statistics shown in

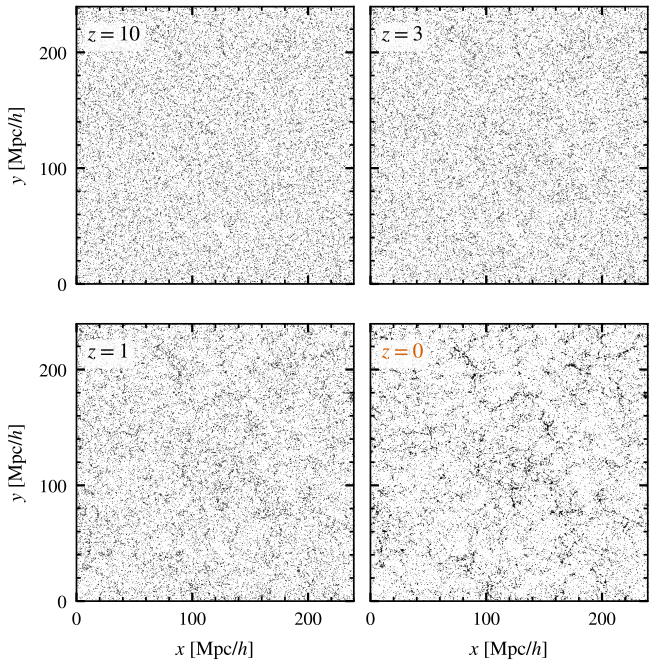


Figure 1. Particles in a 30 Mpc/h thick slab through the whole simulation box ($239.5 \times 239.5\,\mathrm{Mpc}^2/h^2$) at $z = 10$, 3, 1 and 0, from a random 1-in-64 subsample of the particles: the initially uniform matter distribution develops filaments and haloes with cosmic time.

Table 1. Parameters of the Virgo SCDM simulation [30, 31].

Parameter	Symbol	Value
Run		SCDM1
Matter density	Ω_0	1.0
Cosmological constant	Λ	0.0
Hubble parameter	h	0.5
Shape parameter	Γ	0.5
Normalization	σ_8	0.51
Box size	L	239.5 Mpc/h
Particles	N	256^3
Particle mass	m_p	$2.27 \times 10^{11}\,M_\odot/h$
Force softening	l_{soft}	36 kpc/h

this paper are tabulated on a uniform grid of separations with spacing $\Delta r = 1\,\mathrm{kpc}/h$, from $r = 1\,\mathrm{kpc}/h$ to $r \approx 120\,\mathrm{Mpc}/h$ (half the box size), at ten redshifts between $z = 10$ and $z = 0$.

Finally, two relevant datasets from this N-body simulation, i.e., halo-based and correlation-based statistics of dark matter flow, can be found at Zenodo.org [34, 35], along with the accompanying presentation slides, "A comparative study of dark matter flow & hydrodynamic turbulence and its applications" [36]. All data files are also available on GitHub [37].

3 FUNDAMENTAL KINEMATIC RELATIONS

To describe the random and multiscale nature of self-gravitating collisionless dark matter flow (SG-CFD), various statistical measures can be introduced to characterize the density, velocity, and potential fields. There are restrictions on these quantities due to the symmetry implied by the assumptions of homogeneity and isotropy. Based on these assumptions, kinematic relations between different statistical measures can be rigorously developed for a given nature of flow, i.e. incompressible, constant divergence, or irrotational flow.

In this section, we focus on the statistical measures of the velocity field. Three types of statistical quantities will be introduced that are related to each other and are traditionally used to characterize the turbulent flow, i.e. the correlation functions, structure functions, and power spectrum functions. The real-space correlation functions are the most fundamental quantity and building blocks of statistical theory for any stochastic flow. The structure functions and the power spectrum are closely related to correlation functions to describe how the system energy is distributed and transferred across different scales. This section will introduce and review these fundamental statistical quantities and their relations with each other. This will prepare us for better understanding and interpreting the data and results from N-body simulations.

For simplicity, we restrict our discussion to homogeneous and isotropic flow with translational and rotational symmetry in space, respectively. The assumption of homogeneity and isotropy will greatly simplify the form of velocity correlation tensors and facilitate the development of statistical theory. The SG-CFD flow is not required to be stationary (translational symmetry in time), with flow dynamically evolving in the course of time. These statistical quantities are first introduced for the generic flow. Their properties for a given type of flow (i.e. incompressible, constant divergence, and irrotational flow) are presented subsequently. This is important because we will demonstrate that the cosmic peculiar velocity is of constant divergence and irrotational nature on small and large scales (Section 4), respectively.

The starting point is to define a one-dimensional RMS (root mean square) velocity (or velocity dispersion) u:

$$u(a) = \left(\frac{1}{3} \langle \mathrm{u}(\mathrm{x}) \cdot \mathrm{u}(\mathrm{x}) \rangle \right)^{1/2}, \tag{1}$$

where $\langle\rangle$ denotes the average over all particles in system. Here x is the coordinates of a particle. The RMS velocity u is a measure of the mean specific kinetic energy (per unit mass). N-body simulation used here gives $u_0 = u(a = 1) = 354.61 km/s$. The time variation of $u^2(a)$ can be found in Fig. 21 with $u^2(a) \propto t$ once the statistically steady state is established (the simulation gives $u^2 \propto a^{1.23} \propto t^{0.82}$ for $a \geqslant 0.5$; Fig. 21). This leads to a constant rate of the energy cascade $\varepsilon_u \propto u^2/t \approx 4.6 \times 10^{-7} m^2/s^3$, a fundamental constant in the theory of mass and energy cascade [38].

3.1 Two-point first order velocity correlation tensor

We first define a correlation tensor between a velocity field $\mathrm{u}(\mathrm{x})$ and a scalar field $p(\mathrm{x})$. Examples are correlations between the velocity field and the pressure, density, or gravitational potential fields at two different locations x and $\mathrm{x}' = \mathrm{x} + \mathrm{r}$ that are separated by a vector r. The most general form of this tensor in the index notation reads

$$Q_i(\mathrm{x}, \mathrm{r}) = \left\langle u_i(\mathrm{x})\, p\left(\mathrm{x}'\right) \right\rangle. \tag{2}$$

For homogeneous flow, all statistical quantities should be, by definition, independent of location x. The translational symmetry associated with spatial homogeneity leads to $Q_i(\mathrm{x}, \mathrm{r}) \equiv Q_i(\mathrm{r})$. The isotropy symmetry further reduces this tensor to:

$$Q_i(\mathrm{r}) \equiv Q_i(r) = A_1(r)\, r_i, \tag{3}$$

where $A_1(r)$ is a symmetric function of r ($A_1(r)$=$A_1(-r)$) and r_i is the Cartesian components of vector $\boldsymbol{r}$. The divergence of two-point

Table N. Principal notation. Primes denote quantities at $\mathbf{x}' = \mathbf{x} + \mathbf{r}$; $\langle\cdot\rangle$ is an average over all particle pairs with separation r. Velocities are peculiar velocities; lengths are comoving.

Symbol	Meaning	Symbol	Meaning
u, u_0	1D rms peculiar velocity at a and at $a=1$ (Eq. 1)	S_2^{lp}, S_2^{l}	longitudinal structure functions (Eqs. 53, 55)
u_L, $\mathbf{u}_T$	longitudinal and transverse velocity (Eq. 19)	S_2^{ip}, S_2^{i}	total structure functions (Eqs. 66, 67)
Q_i, Q_{ij}	first- and second-order correlation tensors	S_2^{x}	enstrophy structure function (Eq. 72)
R_2, L_2, T_2	total, longitudinal, transverse velocity correlation	E_{nr}	real-space enstrophy distribution (Eq. 78)
R_{2i}	volume average of R_2 (Eq. 52)	R_θ, R_ω	divergence and vorticity correlations
ρ_L, ρ_T	velocity correlation coefficients (Eq. 20)	ξ, $\bar{\xi}$	density correlation and its volume average
$E_u(k)$	velocity (energy) spectrum	$P_\delta(k)$	matter power spectrum
$\sigma_u^2(r)$	energy in scales $> r$ (bulk-flow variance, Eq. 27)	r_t	$L_2 = T_2$ crossover ($\approx 1.3\,\mathrm{Mpc}/h$)
$\sigma_d^2(r)$	energy in scales $< r$ (velocity dispersion, Eq. 29)	r_{t2}	scale with $\sigma_u^2 = \sigma_d^2$
$E_{ur}(r)$	real-space energy distribution (Eq. 32)	r_2	exponential scale of T_2 (Eq. 110)
$a_0(a)$	amplitude of T_2 on large scales	r_1, n	scale and exponent of S_2^l (Eq. 137)
a, H	scale factor, Hubble parameter	r_d	scale of maximum $\langle u_L^2\rangle$

first order correlation tensor $Q_i(r)$ reads

$$\nabla\cdot\mathrm{Q}(\mathrm{x},\mathrm{r}) = \left\langle \frac{\partial u_i(\mathrm{x})}{\partial x_i} p\left(\mathrm{x}^{'}\right)\right\rangle = \frac{\partial Q_i(\mathrm{x},\mathrm{r})}{\partial x_i} = -\frac{\partial Q_i(r)}{\partial r_i}, \tag{4}$$

where the average operator and differential operator can commute $\nabla\cdot\langle\bullet\rangle = \langle\nabla\cdot\bullet\rangle$. Due to $\mathrm{r} = \mathrm{x}^{'} - \mathrm{x}$, we should have

$$\partial/\partial x_i = -\partial/\partial r_i \quad \text{and} \quad \partial/\partial x_i^{'} = \partial/\partial r_i. \tag{5}$$

With Eqs. (3) and (5), the divergence of $Q_i(r)$ (Eq. (4)) reduces to

$$\frac{\partial Q_i(r)}{\partial r_i} = A_1 r_{,i} + \frac{\partial A_1}{\partial r}\frac{\partial r}{\partial r_i} r_i = 3A_1 + \frac{\partial A_1}{\partial r} r. \tag{6}$$

Similarly, the curl of two-point first order correlation tensor is

$$\begin{aligned}\nabla\times\mathrm{Q}(\mathrm{x},\mathrm{r}) &= \varepsilon_{ijk}\frac{\partial Q_i(\mathrm{x},\mathrm{r})}{\partial x_k} = -\varepsilon_{ijk}\frac{\partial Q_i(r)}{\partial r_k} \\ &= -\varepsilon_{ijk}\left(A_1\delta_{ik} + \frac{r_i r_k}{r}\frac{\partial A_1}{\partial r}\right) = 0,\end{aligned} \tag{7}$$

where ε_{ijk} is the Levi-Civita symbol satisfying the identity

$$\varepsilon_{ijk}\delta_{jk} = 0 \quad \text{and} \quad \varepsilon_{ijk} r_j r_k = \mathrm{r}\times\mathrm{r} = 0. \tag{8}$$

The velocity field of self-gravitating collisionless dark matter flow (SG-CFD) can be very complex. We demonstrated that on small scales, virialized haloes are spherical and isotropic. The peculiar velocity of small haloes should have a constant divergence of $-3Ha$. On large scales, velocity is proportional to the gradient of the gravitational potential, curl-free, and irrotational (Zeldovich approximation [39]). Next, we will extend the discussion to three different types of flow.

3.1.1 Results for incompressible flow

The incompressible flow requires $\nabla\cdot\mathrm{u} = 0$ such that the divergence of the correlation tensor vanishes everywhere (Eqs. (4) and (6)) , which leads to $A_1(r) = 0$ or $A_1(r) \sim r^{-3}$. Since $A_1(r)$ must be symmetric with r and regular at origin r=0, $A_1(r) = 0$ must be satisfied for incompressible flow. An important conclusion is that $Q_i(r)$ must be zero for an incompressible, homogeneous, isotropic velocity field.

3.1.2 Results for constant divergence flow

For flow with constant divergence everywhere such that $\nabla\cdot\mathrm{u} = \theta$. This leads to a constant divergence for the correlation tensor (Eq. (6)),

$$\frac{\partial Q_i(r)}{\partial r_i} = 3A_1 + \frac{\partial A_1}{\partial r} r = -\theta\langle p(\mathrm{x})\rangle \tag{9}$$

such that solution $A_1(r) = -\theta\langle p(\mathrm{x})\rangle/3$ is independent of r.

3.1.3 Results for irrotational flow

Irrotational flow requires curl free for velocity field ($\nabla\times\mathrm{u} = 0$). The curl of $Q_i(r)$ vanishes, i.e. $\nabla\times\mathrm{Q}(\mathrm{x},\mathrm{r}) \equiv 0$ (Eq. (7)).

3.2 Two-point second order velocity correlation tensor

Physically, a two-point velocity correlation asks: if one particle moves in some direction, how likely is a second particle a distance r away to move the same way? Splitting each velocity into its component along the line joining the pair (longitudinal) and perpendicular to it (transverse) separates converging or diverging motion of the pair from shearing motion; the kinematic constraints of each flow type then tie the two components together. The two-point second order velocity correlation tensor quantifies the degree to which the velocities at two locations are statistically correlated. It can be defined as (similarly, the dependence on location $\boldsymbol{x}$ and direction $\boldsymbol{r}$ is dropped due to homogeneity and isotropy)

$$Q_{ij}(\mathrm{r}) = Q_{ij}(r) = \left\langle u_i(\mathrm{x}) u_j\left(\mathrm{x}^{'}\right)\right\rangle. \tag{10}$$

In the N-body simulation, this average $\langle\cdot\rangle$ is taken over **all pairs** of particles with the same separation r (Fig. 2). A more compact alternative notation is to write $Q_{ij}(r) \equiv \left\langle u_i u_j^{'}\right\rangle$, where the prime indicates that quantity is evaluated at $\mathrm{x}^{'} = \mathrm{x}+\mathrm{r}$. Due to symmetry in Eq. (10)

$$Q_{ij}(-r) = \left\langle u_i^{'} u_j\right\rangle = Q_{ji}(r) = Q_{ij}(r) = \left\langle u_i u_j^{'}\right\rangle. \tag{11}$$

Einstein index notation is adopted hereafter that implies the summation over a specific index,

$$Q_{ii}(r) = \left\langle u_i u_i^{'}\right\rangle = \left\langle u_1 u_1^{'}\right\rangle + \left\langle u_2 u_2^{'}\right\rangle + \left\langle u_3 u_3^{'}\right\rangle. \tag{12}$$

The most general form of $Q_{ij}(r)$ reads

$$Q_{ij}(\mathrm{r}) = Q_{ij}(r) = A_2(r) r_i r_j + B_2(r)\delta_{ij}, \tag{13}$$

where $A_2(r)$ and $B_2(r)$ are scalar functions of r only. Finally, the divergence of $Q_{ij}(r)$ can be derived from Eq. (13)

$$Q_{ij,i} = \frac{\partial Q_{ij}(r)}{\partial r_i} = \left(4A_2 + \frac{\partial A_2}{\partial r} r + \frac{1}{r}\frac{\partial B_2}{\partial r}\right) r_j. \tag{14}$$

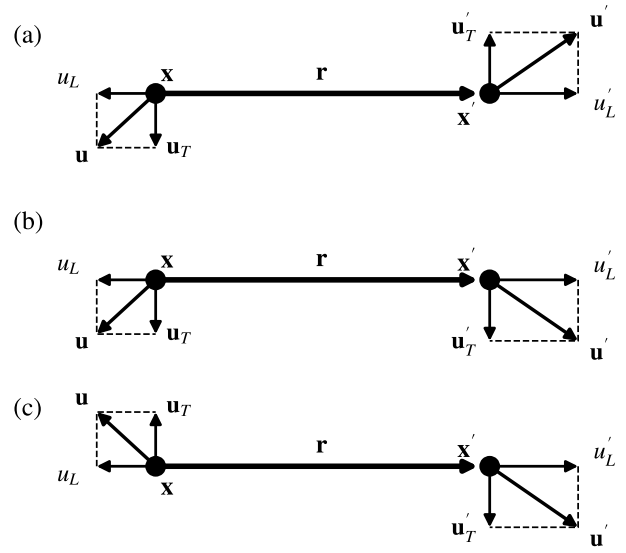


Figure 2. Geometry of a particle pair. Particles at $\mathbf{x}$ and $\mathbf{x}' = \mathbf{x} + \mathbf{r}$ have velocities $\mathbf{u}$ and $\mathbf{u}'$, each split into a longitudinal component (u_L, u'_L) along $\mathbf{r}$ and a transverse component ($\mathbf{u}_T$, $\mathbf{u}'_T$) perpendicular to it (Eq. 19). Panels (a)–(c) show three sign combinations: in (a) the longitudinal components point apart and the transverse components are antiparallel; in (b) the longitudinal components point apart and the transverse components are parallel; in (c) the transverse components are antiparallel.

The curl of second order correlation tensor is derived as

$$\nabla \times Q_{ij}(r) = \varepsilon_{imk} Q_{mj,k} = \left(A_2 \varepsilon_{imj} r_m + \frac{1}{r}\frac{\partial B_2}{\partial r} \varepsilon_{ijk} r_k \right) = \varepsilon_{imj} r_m \left(A_2 - \frac{1}{r}\frac{\partial B_2}{\partial r} \right). \tag{15}$$

3.2.1 *Second order velocity correlation functions*

It is more convenient to introduce three scalar velocity correlation functions by index contraction of $Q_{ij}(r)$ in Eq. (13). Namely, a total correlation function $R_2(r)$ can be defined as

$$R_2(r) = Q_{ij}\delta_{ij} = \left\langle \mathrm{u} \cdot \mathrm{u}' \right\rangle = \left\langle u_i u'_i \right\rangle = A_2 r^2 + 3B_2. \tag{16}$$

The longitudinal correlation function $L_2(r)$ is defined as

$$L_2(r) = Q_{ij} r_i r_j / r^2 = \left\langle u_L u'_L \right\rangle = A_2 r^2 + B_2, \tag{17}$$

where $u_L = \mathrm{u} \cdot \hat{\mathrm{r}} = u_i \hat{r}_i$ is the longitudinal velocity and $\hat{r}_i = r_i / r$ is the normalized Cartesian component of vector $\mathbf{r}$ between two particles. The transverse (lateral) correlation function reads:

$$T_2(r) = Q_{ij} n_i n_j = \left\langle \mathrm{u}_T \cdot \mathrm{u}'_T \right\rangle / 2 = B_2(r), \tag{18}$$

where $\boldsymbol{n}$ is normal vector perpendicular to the direction of separation $\mathbf{r}$ ($\mathrm{n} \cdot \hat{\mathrm{r}} = 0$) and

$$\mathrm{u}_T = -(\mathrm{u} \times \hat{\mathrm{r}} \times \hat{\mathrm{r}}) = \mathrm{u} - (\mathrm{u} \cdot \hat{\mathrm{r}})\hat{\mathrm{r}} \tag{19}$$

is the transverse velocity perpendicular to vector $\boldsymbol{r}$ with $\times$ standing for the cross product. Figure 2 provides a sketch of the longitudinal and transverse velocities.

Two correlation coefficients can be defined for longitudinal and transverse velocity,

$$\rho_L(r) = \frac{\left\langle u_L u'_L \right\rangle}{\left\langle u_L^2 \right\rangle} \quad \text{and} \quad \rho_T(r) = \frac{\left\langle \mathrm{u}_T \cdot \mathrm{u}'_T \right\rangle}{\left\langle |\mathrm{u}_T|^2 \right\rangle}, \tag{20}$$

both of which should vanish at large distance $r \to \infty$ (the velocities of two particles far from each other should be uncorrelated) and increase with $r \to 0$ due to the increasing gravitational interaction (Fig. 17). Similarly, we can define the angles between two longitudinal and transverse velocities:

$$\langle \cos(\theta_L) \rangle = \left\langle \frac{u_L u'_L}{|u_L|\,|u'_L|} \right\rangle, \quad \langle \cos(\theta_T) \rangle = \left\langle \frac{\mathrm{u}_T \cdot \mathrm{u}'_T}{|\mathrm{u}_T|\,|\mathrm{u}'_T|} \right\rangle. \tag{21}$$

The angle between velocity vector u and $\boldsymbol{r}$ vector and the angle between two velocities u and u′ can also be defined for all pair of particles with the same separation r,

$$\langle \cos(\theta_{\mathrm{ur}}) \rangle = \left\langle \frac{\mathrm{u} \cdot \mathrm{r}}{|\mathrm{u}|\,|\mathrm{r}|} \right\rangle, \quad \langle \cos(\theta_{\mathrm{uu}'}) \rangle = \left\langle \frac{\mathrm{u} \cdot \mathrm{u}'}{|\mathrm{u}|\,|\mathrm{u}'|} \right\rangle. \tag{22}$$

Note that all averaging is performed over all pairs of particles with the same separation r. These quantities will greatly improve our understanding of the velocity field (Figs. 17, 18, 19, and 20). The total velocity correlation reads (Eqs. (16)-(18))

$$R_2(r) = 2R(r) = L_2(r) + 2T_2(r), \tag{23}$$

where $R(r)$ is the standard velocity correlation function in turbulence literature. Three second order correlation functions from N-body simulations are presented in Figs. 3, 4, and 5.

The velocity power spectrum $E_u(k)$ and correlation function $R(r)$ are related to each other through the Fourier transform pair

$$R(r) = \int_0^\infty E_u(k) \frac{\sin(kr)}{kr} dk, \tag{24}$$

$$E_u(k) = \frac{2}{\pi} \int_0^\infty R(r)\, kr \sin(kr)\, dr. \tag{25}$$

A length scale l_{u0} (integral scale) is introduced as

$$l_{u0} = \frac{1}{u^2} \int_0^\infty R(r)\, dr = \frac{\pi}{2u^2} \int_0^\infty E_u(k)\, k^{-1} dk \tag{26}$$

that defines a characteristic length scale within which, velocities are appreciably correlated. This is the length scale at which energy is injected into or taken out of the system.

3.2.2 *Second order velocity dispersion functions and energy distribution in real space*

The total kinetic energy can be divided at any scale r into the energy of motions coherent over regions larger than r (bulk flows) and the energy of motions inside regions of size r (random or virial motions). The two dispersion functions below measure these two parts, and their derivative with respect to r shows where in scale the energy resides. The one-dimensional variance of the smoothed velocity (the bulk flow velocity) with a filter of size r can be written as follows:

$$\sigma_u^2(r) = \frac{1}{3} \int_{-\infty}^\infty E_u(k) W(kr)^2 dk = \int_r^\infty E_{ur}\left(r'\right) dr', \tag{27}$$

which is the variance of smoothed velocity field with a filter of size r, or equivalently the kinetic energy contained in all scales above r. Here $E_{ur}(r)$ is the energy density in the real space that describes the energy distribution with respect to scale r, just like the energy

spectrum $E_u(k)$ in Fourier space with respect to the wavenumber k (Fig. 12). Here $W(x)$ is a window function. For a top-hat spherical filter,

$$W(x) = \frac{3}{x^3}\left[\sin(x) - x\cos(x)\right] = 3\frac{j_1(x)}{x}, \tag{28}$$

where $j_1(x)$ is the *first* order spherical Bessel function of the *first* kind. The velocity dispersion on the same scale r is

$$\sigma_d^2(r) = \frac{1}{3}\int_{-\infty}^{\infty} E_u(k)\left[1 - W(kr)^2\right]dk, \tag{29}$$

which is the kinetic energy contained in scales below r.

The exact relation between the total velocity correlation $R_2(r)$ and $\sigma_u^2(r)$ can be derived from Eqs. (24) and (27)

$$R_2(2r) = \frac{1}{24r^2}\frac{\partial}{\partial r}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^3\frac{\partial}{\partial r}\left(\sigma_u^2(r)\,r^4\right)\right)\right). \tag{30}$$

The total kinetic energy is decomposed into two parts:

$$\sigma_u^2(r) + \sigma_d^2(r) = u^2. \tag{31}$$

The specific kinetic energy of entire system u^2 can have two separate contributions on any scale of r, i.e. the velocity dispersion $\sigma_d^2(r)$ for kinetic energy contained in scales below r and velocity variance $\sigma_u^2(r)$ for kinetic energy contained in scales above r (Fig. 10). From Eq. (30), total velocity correlation $R_2(r)$ can be used to derive both dispersion functions.

Just like the energy spectrum $E_u(k)$ in a Fourier space, the energy distribution in the real space with respect to the scale r can be found from dispersion functions as (from Eq. (27)),

$$E_{ur}(r) = -\frac{\partial\sigma_u^2(r)}{\partial r}. \tag{32}$$

The connection with energy spectrum $E_u(k)$ in Fourier space can be found from Eq. (27),

$$E_{ur}(r)\,r^2 = -\frac{4}{3}\int_0^{\infty} E_u\left(\frac{x}{r}\right)W(x)\,W'(x)\,x dx, \tag{33}$$

where $x = kr$ and $W'(x)$ is the derivative of $W(x)$ with respect to x. In principle, $E_{ur}(r)$ and $E_u(k)$ contain the same information about the distribution of energy.

Two velocity dispersion functions $\sigma_d^2(r)$ and $\sigma_u^2(r)$ can be derived from the velocity correlation function $R_2(r)$, which in turn can be obtained accurately from the N-body simulation (Fig. 15). Specifically, for a top-hat filter, a power-law spectrum $E_u(k) \equiv bk^{-m}$ will lead to (from Eqs. (24) and (27)),

$$R_2(r) = -2b\Gamma(-m)\sin\left(\frac{m\pi}{2}\right)r^{m-1}, \tag{34}$$

$$\begin{aligned}\sigma_d^2(r) = &-6\cdot 2^{2+m}(1+m)(4+m)\\ &\cdot\Gamma(-5-m)\sin\left(\frac{m\pi}{2}\right)br^{m-1},\end{aligned} \tag{35}$$

where two functions satisfy relation in Eq. (30). For self-gravitating collisionless dark matter flow,

$$\sigma_d^2(r) = \left(\sigma_{11}^2(r) + \sigma_{22}^2(r) + \sigma_{33}^2(r)\right)/3, \tag{36}$$

where $\sigma_{ij}^2(r)$ is the velocity dispersion tensor on scale of r. Off-diagonal terms are small and negligible (Fig. 16).

3.2.3 Results for incompressible flow

The divergence-free condition for Q_{ij} leads to (Eq. (14))

$$r\frac{\partial B_2}{\partial r} = \frac{1}{r}\frac{\partial\left(r^2B_2\right)}{\partial r} - 2B_2 = -\frac{1}{r}\frac{\partial\left(r^4A_2\right)}{\partial r}. \tag{37}$$

Equation (37) can be rewritten as

$$B_2 = \frac{1}{2r}\frac{\partial\left(r^4A_2 + r^2B_2\right)}{\partial r}. \tag{38}$$

With the help of Eqs. (17), (18) and (23), the kinematic relations between three scalar correlations functions read

$$T_2 = \frac{1}{2r}\left(r^2L_2\right)_{,r} \quad \text{and} \quad R_2 = \frac{1}{r^2}\left(r^3L_2\right)_{,r}. \tag{39}$$

The two scalar functions $A_2(r)$ and $B_2(r)$, and the correlation tensor $Q_{ij}(r)$ can be finally expressed in terms of the longitudinal correlation function $L_2(r)$ as

$$A_2 = -\frac{1}{2r}(L_2)_{,r} \quad \text{and} \quad B_2 = \frac{1}{2r}\left(r^2L_2\right)_{,r}, \tag{40}$$

$$Q_{ij}(r) = -\frac{1}{2r}\left[(L_2)_{,r}r_ir_j - \left(r^2L_2\right)_{,r}\delta_{ij}\right]. \tag{41}$$

The relations between the longitudinal $L_2(r)$, transverse correlation functions $T_2(r)$, and velocity power spectrum $E_u(k)$ can be equivalently derived using Eqs. (25) and (39)

$$L_2(r) = \int_0^{\infty} E_u(k)\frac{2j_1(kr)}{kr}dk, \tag{42}$$

$$T_2(r) = \int_0^{\infty} E_u(k)\left(j_0(kr) - \frac{j_1(kr)}{kr}\right)dk, \tag{43}$$

where $j_n(kr)$ is the nth order spherical Bessel function of the *first* kind. The first two spherical Bessel functions are

$$\begin{aligned}j_0(x) &= \frac{\sin(x)}{x} \approx 1 - \frac{x^2}{6},\\ j_1(x) &= \frac{\sin(x)}{x^2} - \frac{\cos(x)}{x} \approx \frac{x}{3} - \frac{x^3}{30}.\end{aligned} \tag{44}$$

With Eqs. (39) and (42) and integration by parts, the integral length scale l_{u0} for incompressible flow is

$$\begin{aligned}l_{u0} &= \frac{1}{u^2}\int_0^{\infty} R(r)\,dr = \frac{1}{u^2}\int_0^{\infty} L_2(r)\,dr,\\ \int_0^{\infty} T_2(r)\,dr &= \frac{1}{2}\int_0^{\infty} L_2(r)\,dr.\end{aligned} \tag{45}$$

3.2.4 Results for constant divergence flow

For constant divergence flow with $\nabla\cdot\mathrm{u} = \theta$, we should have

$$Q_{ij,i} = \frac{\partial Q_{ij}(r)}{\partial r_i} = -\left\langle\frac{\partial u_i}{\partial x_i}u_j'\right\rangle = -\theta\left\langle u_j'\right\rangle = \theta\left\langle u_j\right\rangle. \tag{46}$$

It can be shown that $Q_{ij,i} = 0$ for velocity field with a zero mean ($\langle u_j'\rangle = 0$) such that Eq. (37) for incompressible flow is still valid for constant divergence flow. Therefore, all results for second order correlation functions in Section 3.2.3 are also valid for constant divergence flow. More generally, the incompressible and constant divergence flows share the same kinematic relations on even order. However, kinematic relations can be different in odd order for incompressible and constant divergence flows [29].

3.2.5 Results for irrotational flow

The curl-free condition for irrotational flow (Eq. (15)) leads to different kinematic relations between correlation functions,

$$R_2 = \frac{1}{r^2}\left(r^3 T_2\right)_{,r} \quad \text{and} \quad L_2 = \left(rT_2\right)_{,r}. \tag{47}$$

Similarly, the velocity correlation tensor $Q_{ij}(r)$ can be written in terms of the transverse correlation function $T_2(r)$ only

$$Q_{ij}(r) = (T_2)_{,r}\frac{r_i r_j}{r} + T_2\delta_{ij}. \tag{48}$$

Relations between longitudinal and transverse correlation functions and the velocity power spectrum $E_u(k)$ can be found as (from Eq. (47)) or see Gorski [40]

$$L_2(r) = 2\int_0^\infty E_u(k)\left(j_0(kr) - 2\frac{j_1(kr)}{kr}\right)dk, \tag{49}$$

$$T_2(r) = \int_0^\infty E_u(k)\frac{2j_1(kr)}{kr}dk. \tag{50}$$

The integral length scale l_{u0} for irrotational velocity reads

$$\begin{aligned} l_{u0} &= \frac{1}{u^2}\int_0^\infty R(r)\,dr = \frac{1}{u^2}\int_0^\infty T_2(r)\,dr, \\ &\int_0^\infty L_2(r)\,dr = 0. \end{aligned} \tag{51}$$

3.2.6 The nature of dark matter flow on small and large scales

Clearly, different kinematic relations between correlation functions (Eqs. (39) and (47)) can be used to characterize the nature of flow (flow characterization). For example, we can compute the integral $R_{2i}(r)$ of total velocity correlation function $R_2(r)$ as

$$R_{2i} = \frac{1}{r^3}\int_0^r R_2(y)y^2 dy. \tag{52}$$

It can be shown that $R_{2i} = L_2$ (from Eq. (39)) if the flow is incompressible (or constant divergence flow) and $R_{2i} = T_2$ (from Eq. (47)) if the flow is irrotational. This information can be used to demonstrate that the self-gravitating collisionless dark matter flow is of constant divergence on small scale and irrotational on large scale (see Fig. 6).

3.3 Second order velocity structure functions

Structure functions are common statistical measures for turbulent flow. Where a correlation function measures how alike two velocities are, a structure function measures how different they are: the mean square velocity difference across a separation r, i.e. the energy of relative motion on scales below r. We will introduce different structure functions as quantitative measures for kinetic energy and enstrophy contained in scales below r. Structure functions are relatively easy to measure from observation.

3.3.1 Second order longitudinal structure function and limiting correlations on small scale

The most common one is the second order longitudinal structure function originally defined as

$$S_2^{lp}(r) = \left\langle (\Delta u_L)^2 \right\rangle = \left\langle \left(u_L' - u_L\right)^2 \right\rangle = 2\left(\left\langle u_L^2 \right\rangle - L_2(r)\right), \tag{53}$$

where $\left\langle u_L^2 \right\rangle$ is the second moment of longitudinal velocity u_L on scale r (Fig. 24). The difference in velocity on scale r (or pairwise velocity in cosmology) is defined as $\Delta u_L(r) = u_L' - u_L$. The first order longitudinal structure function reads

$$S_1^{lp}(r) = \langle \Delta u_L \rangle = \left\langle u_L' - u_L \right\rangle, \tag{54}$$

which is essentially the mean pairwise velocity.

A modified (alternative) definition of second order longitudinal structure function is

$$S_2^{l}(r) = 2\left(\lim_{r\to 0}\left\langle u_L u_L' \right\rangle - L_2(r)\right) = 2\left(u^2 - L_2(r)\right), \tag{55}$$

where u^2 is one-dimensional velocity dispersion in Eq. (1). Two structure functions $S_2^{lp}(r) \neq S_2^{l}(r)$ are a manifestation of the collisionless nature of dark matter flow.

Remarks: For collisional incompressible flow $\left\langle u_L^2 \right\rangle = u^2$ on all scales, where $\left\langle u_L^2 \right\rangle$ is independent of the scale r. In addition

$$\lim_{r\to 0} L_2(r) = \lim_{r\to 0}\left\langle u_L u_L' \right\rangle = u^2 = \lim_{r\to 0}\left\langle u_L^2 \right\rangle, \tag{56}$$

where u_L and u_L' are fully correlated when $r \to 0$ due to the collisional interaction. Or equivalently, the correlation coefficient $\rho_L(r=0) = 1$ (see Eq. (20)). For incompressible flow, two definitions are equivalent, that is, $S_2^{lp}(r) = S_2^{l}(r)$ and $S_2^{lp}(0) = S_2^{l}(0) = 0$.

However, this picture is completely different for dark matter flow (SG-CFD). Due to the collisionless nature, the longitudinal velocities u_L and u_L' are not fully correlated on small scales ($r \to 0$). The maximum correlation coefficient between u_L and u_L' is $\rho_L(r=0) = 0.5$ that can be estimated as follows:

According to the description of the halo of the self-gravitating system, any pair of particles with small separation $r \to 0$ should come from the same halo, while different pairs can come from different haloes of different sizes. The particle velocity dispersion in haloes can be decomposed into

$$\begin{aligned} \sigma^2(m_h) &= \sigma_v^2(m_h) + \sigma_h^2(m_h) \\ &= \left(1 - \rho_{cor}(m_h)\right)\sigma^2 + \rho_{cor}(m_h)\,\sigma^2, \end{aligned} \tag{57}$$

i.e. the sum of halo velocity dispersion σ_h^2 and halo virial dispersion σ_v^2, where m_h is the mass of the halo that the particle resides in. The correlation coefficient ρ_{cor} between the velocities of the pair of particles in haloes of mass m_h approximately reads,

$$\rho_{cor}(m_h) = \frac{\sigma_h^2}{\sigma^2} = \frac{1}{1 + \sigma_v^2/\sigma_h^2}. \tag{58}$$

The velocities of the pair of particles with separation $r \to 0$ in small haloes should be fully correlated, that is, $\rho_{cor}\left(m_h \ll m_h^*\right) \approx 1$ with $\sigma_v^2 \ll \sigma_h^2$ for small haloes. While velocities of particle pairs from large haloes should be independent of each other, that is, $\rho_{cor}\left(m_h \gg m_h^*\right) \approx 0$ with $\sigma_v^2 \gg \sigma_h^2$. Here m_h^* is a characteristic mass scale.

The average correlation for all pairs with the same r from all haloes of different size can be estimated as $\sigma_v^2 = \sigma_h^2$ such that $\langle \rho_{cor} \rangle = 0.5$ when $r \to 0$ (see Fig. 17 from the N-body simulation). The average correlation coefficient of the longitudinal velocity for all pairs on the limiting scale $r \to 0$ is $\rho_L(r=0) = \langle \rho_{cor} \rangle = 0.5$ such that

$$\lim_{r\to 0} L_2(r) = \lim_{r\to 0}\left\langle u_L u_L' \right\rangle = \lim_{r\to 0}\rho_L\left\langle u_L^2 \right\rangle = \frac{1}{2}\lim_{r\to 0}\left\langle u_L^2 \right\rangle. \tag{59}$$

Therefore, in SG-CFD, two structure functions are not equivalent,

i.e. $S_2^{lp}(r) \neq S_2^{l}(r)$ (Eqs. (53) and (55)). On the smallest scale $r \to 0$, the original definition

$$S_2^{lp}(r=0) = \lim_{r\to 0}\left\langle u_L^2 \right\rangle \quad \text{and} \quad S_2^{l}(r=0) = 0, \tag{60}$$

while on the largest scale (see Fig. 15)

$$S_2^{lp}(r=\infty) = S_2^{l}(r=\infty) = 2u^2. \tag{61}$$

3.3.2 *Mass-energy and momentum conservation for collisionless "annihilation" at $r = 0$*

The fact that $\rho_L(r=0) \neq 1$ might lead to some interesting phenomena on the smallest scale. Just like the electron-positron annihilation, let us check the simplest "one-dimensional annihilation" scenario of two dark matter particles on the smallest scale $r = 0$. Assuming that gravity is the only interaction and no radiation is produced from "annihilation", two particles with the same mass m only annihilate when the separation between them vanishes ($r = 0$). The conservation of momentum requires

$$mv_1 + mv_2 = m'v_3 \quad \text{such that} \quad v_3 = \frac{m}{m'}(v_1 + v_2), \tag{62}$$

where m' and v_3 are the rest mass and velocity of new particle after "annihilation". Taking the average for all pairs of particles with $r = 0$, $v_1 = u_L$, and $v_2 = u_L^{'}$

$$\begin{aligned}\left\langle v_3^2 \right\rangle = \left(\frac{m}{m'}\right)^2 \left\langle (v_1+v_2)^2 \right\rangle &= 2\left(\frac{m}{m'}\right)^2 \left(\left\langle v_1^2 \right\rangle + \left\langle v_1 v_2 \right\rangle\right) \\ &= 2\left(\frac{m}{m'}\right)^2 \left\langle u_L^2 \right\rangle (1+\rho_{L0}),\end{aligned} \tag{63}$$

where $\rho_{L0} \equiv \rho_L(r=0)$ is the limiting velocity correlation coefficient at $r = 0$. Assuming no radiation is produced, the conservation of total mass-energy can be written as

$$mc^2 + \frac{1}{2}mv_1^2 + mc^2 + \frac{1}{2}mv_2^2 = m'c^2 + \frac{1}{2}m'v_3^2, \tag{64}$$

where c is the speed of light. Taking average and we have:

$$\begin{aligned}m' &= m\left[1 + \frac{\left\langle u_L^2 \right\rangle}{2c^2} + \sqrt{1 - \rho_{L0}\frac{\left\langle u_L^2 \right\rangle}{c^2} + \frac{\left\langle u_L^2 \right\rangle^2}{4c^4}}\right] \\ &\approx m\left[2 + (1-\rho_{L0})\frac{\left\langle u_L^2 \right\rangle}{2c^2}\right].\end{aligned} \tag{65}$$

Therefore, for $\rho_{L0} = 1/2 \neq 1$, the particle "annihilation" leads to extra mass converted from the kinetic energy if gravity is the only interaction and no radiation is produced. If the mass is conserved ($m^{'} = 2m$), the total radiation should be produced on the order of $M\langle u_L^2 \rangle$, where M is the total mass of dark matter [41].

3.3.3 *Second order total structure function for kinetic energy*

The second order total velocity structure functions are introduced in the vector forms,

$$S_2^{ip}(r) = \left\langle \Delta \mathrm{u}^2 \right\rangle = \left\langle \left(\mathrm{u}^{'} - \mathrm{u}\right)^2 \right\rangle = 2\left\langle |\mathrm{u}|^2 \right\rangle - 2R_2(r), \tag{66}$$

$$S_2^{i}(r) = 6u^2 - 2R_2(r), \tag{67}$$

where $S_2^{ip}(r)$ is the original version and $S_2^{i}(r)$ is the modified version; here $\langle |\mathbf{u}|^2 \rangle$ is averaged over the pairs at separation r, and it equals $3\langle u_L^2 \rangle$ only if the pair velocities are isotropic, which the simulation violates by 10–15 per cent at $0.1 < r < 5\,\mathrm{Mpc}/h$. Just like the longitudinal structure functions in Section 3.3.1, two definitions are equivalent for incompressible flow and $S_2^{i}(r) \neq S_2^{ip}(r)$ for SG-CFD. The relation between the structure function $S_2^{i}(r)$ and the velocity spectrum $E_u(k)$ can be found from Eq. (24)

$$S_2^{i}(r) = 4\int_0^\infty E_u(k)\left(1 - j_0(kr)\right)dk, \tag{68}$$

where $S_2^{i}(r=0) = 0$, just like the modified longitudinal structure function $S_2^{l}(r=0) = 0$.

For a power-law velocity spectrum $E_u(k) \equiv bk^{-m}$, we have a power-law structure function:

$$\begin{aligned}S_2^{i}(r) &= 4\int_0^\infty E_u(k)\left[1 - j_0(kr)\right]dk \\ &= -\frac{2^{2-m}\Gamma(3/2)\Gamma((1-m)/2)}{\Gamma((m+2)/2)}br^{m-1}.\end{aligned} \tag{69}$$

The exact relation between the total structure function $S_2^{i}(r)$ and the one-dimensional velocity dispersion $\sigma_d^2(r)$ can be derived using Eqs. (67), (31) and (30)

$$S_2^{i}(2r) = \frac{1}{12r^2}\frac{\partial}{\partial r}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^3\frac{\partial}{\partial r}\left(\sigma_d^2(r)\,r^4\right)\right)\right), \tag{70}$$

where $S_2^{i}(r)$ is a quantity relevant to the specific kinetic energy contained below the scale r (i.e. $\sigma_d^2(r)$ in Eq. (29)) through Eq. (70). While the total second order correlation $R_2(r)$ is related to the specific kinetic energy above the scale r (i.e. $\sigma_u^2(r)$ in Eq. (31)) through Eq. (30).

3.3.4 *Second order structure function for enstrophy and real space enstrophy distribution*

The enstrophy of entire system can be of interest for some applications, where the total enstrophy E_n in the system is defined as

$$E_n = \int_0^\infty E_u(k)\,k^2 dk. \tag{71}$$

Enstrophy is related to the integral of the gradient of the velocity field with contributions from both divergence and vorticity (Eq. (92)).

We can introduce a different structure function that is relevant to enstrophy on different scales. For a top-hat filter of size r, we introduce the structure function $S_2^{x}(r)$ (see Fig. 13 for the simulation result)

$$\begin{aligned}S_2^{x}(r) &= \frac{2}{3}r^2\int_0^\infty E_u(k)\,k^2 W^2(kr)\,dk \\ &= 6\int_0^\infty E_u(k)\,j_1^2(kr)\,dk,\end{aligned} \tag{72}$$

where $W(kr) = 3j_1(kr)/(kr)$ is the window function and r is the radius of spherical filter. The structure function $S_2^{x}(r)$ can be related to the one-dimensional enstrophy of the smoothed velocity field, similar to the definition of velocity dispersion $\sigma_u^2(r)$ in Eq. (27). The following equation shows that $S_2^{x}(r)/\left(2r^2\right)$ is a measure of smoothed one-dimensional enstrophy on the scale of r, or the total enstrophy contained in scales greater than r,

$$\frac{S_2^{x}(r)}{2r^2} = \frac{1}{3}\int_0^\infty E_u(k)\,k^2 W^2(kr)\,dk = \int_r^\infty E_{nr}\left(r^{'}\right)dr^{'} \tag{73}$$

Here $E_{nr}(r)$ is the density of enstrophy in real space. For collisional incompressible flow, the total enstrophy of the entire system

is finite with $S_2^x(r\to 0)\propto r^2$. On the contrary, the total enstrophy in collisionless flow diverges with $r\to 0$ such that the structure function $S_2^x(r)$ should approach zero at a rate slower than r^2 (Eq. (73)). Let us apply a sharp spectral filter in the Fourier space with a cutoff wavenumber K, at which the cutoff enstrophy equals that of the filtered velocity field, namely,

$$\int_0^\infty E_u(k)\,k^2 W(kr)^2\,dk=\int_0^K E_u(k)\,k^2 dk. \tag{74}$$

A relation can be established between the cutoff wavenumber K and scale r from Eq. (73),

$$\frac{3}{2}\frac{S_2^x(r)}{r^2}=\int_0^K E_u(k)\,k^2 dk. \tag{75}$$

With $K\to\infty$, $S_2^x(r)$ must approach zero with a rate slower than r^2 to allow a diverging enstrophy.

The exact relation between two structure functions $S_2^i(r)$ and $S_2^x(r)$ can be obtained from Eqs. (68) and (72)

$$\frac{1}{3r^2}\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}\left(S_2^x(r)\,r^4\right)\right)=\frac{\partial S_2^i(2r)}{\partial r}, \tag{76}$$

where both $S_2^i(r)$ and $S_2^x(r)$ can be found from the total velocity correlation function $R_2(r)$.

The structure function $S_2^x(r)$ has a special property that itself is a measure of the specific kinetic energy on scales below r (see Eq. (76), just like $\sigma_d^2(r)$), while $S_2^x(r)/\left(2r^2\right)$ is a direct measure of the enstrophy contained in the scales above r (Eq. (73) and Fig. 13). Specifically, for the power-law velocity spectrum $E_u(k)\equiv bk^{-m}$,

$$\begin{aligned}S_2^x(r)=&-3\cdot 2^{1+m}(m+2)(m-1)\\&\cdot\Gamma(-3-m)\sin\left(\frac{m\pi}{2}\right)br^{m-1}.\end{aligned} \tag{77}$$

Clearly, both Eqs. (69) and (77) satisfy the relation in Eq. (76). Just like the energy distribution in real space (Eq. (32)), the enstrophy distribution in real space can be obtained from $S_2^x(r)$ as (from Eq. (73) and see Fig. 14)

$$E_{nr}(r)=-\frac{\partial}{\partial r}\left[S_2^x(r)/\left(2r^2\right)\right]. \tag{78}$$

The relation between $E_{nr}(r)$ and the velocity spectrum $E_u(k=x/r)$ is

$$E_{nr}(r)\,r^4=-\frac{2}{3}\int_0^\infty E_u\left(\frac{x}{r}\right)W(x)\,W'(x)\,x^3dx. \tag{79}$$

3.3.5 Results for incompressible flow

Some kinematic relations between second order structure functions can be derived and presented here. First, the longitudinal structure function can be related to the velocity spectrum for incompressible flow using Eqs.(42) and (55)

$$S_2^l(r)=\frac{4}{3}\int_0^\infty E_u(k)\left(1-3\frac{j_1(kr)}{kr}\right)dk. \tag{80}$$

The exact relations with the second order total velocity structure function can be found from Eqs. (68) and (80)

$$S_2^i(r)=\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^3 S_2^l(r)\right]. \tag{81}$$

The exact relation to the velocity dispersion function $\sigma_d^2(r)$ is as follows:

$$S_2^l(2r)=\frac{1}{12r^5}\frac{\partial}{\partial r}\left(r^3\frac{\partial}{\partial r}\left(\sigma_d^2(r)\,r^4\right)\right), \tag{82}$$

where the longitudinal structure function $S_2^l(r)$ is also a quantitative measure of the specific kinetic energy contained in scales below r for incompressible flow. For power-law velocity spectrum $E_u(k)\equiv bk^{-m}$ (from Eq. (80))

$$S_2^l(r)=-\frac{1}{3}\cdot\frac{2^{2-m}\Gamma(1+3/2)\,\Gamma((1-m)/2)}{\Gamma((m+4)/2)}br^{m-1}. \tag{83}$$

Let us assume a power-law scaling for velocity dispersion function $\sigma_d^2(r)\equiv br^n$ that can be inserted into Eqs. (70), (76), and (82) to obtain the three velocity structure functions that follow the same power-law scaling as

$$S_2^l(r)=\frac{(4+n)(6+n)}{12\cdot 2^n}br^n, \tag{84}$$

$$S_2^i(r)=\frac{(3+n)(4+n)(6+n)}{12\cdot 2^n}br^n, \tag{85}$$

$$S_2^x(r)=\frac{n(3+n)(6+n)}{4(2+n)}br^n. \tag{86}$$

We will use these equations in Section 4 to understand the distribution of kinetic energy across different scales in N-body simulations.

3.3.6 Results for constant divergence flow

Just like the correlation functions in Section 3.2, all results for second order structure functions of incompressible flow in Section 3.3.5 are still valid for constant divergence flow.

3.3.7 Results for irrotational flow

Similar relations for the longitudinal structure function can be derive exactly for irrotational flow. The relation to the velocity spectrum is (from Eqs. (49) and (55))

$$S_2^l(r)=\frac{4}{3}\int_0^\infty E_u(k)\left(1-3j_0(kr)+6\frac{j_1(kr)}{kr}\right)dk. \tag{87}$$

The relation with the second order total velocity structure function $S_2^i(r)$ is derived as,

$$\frac{\partial\left[rS_2^i(r)\right]}{\partial r}=\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^3 S_2^l(r)\right]. \tag{88}$$

3.4 Correlation functions for velocity gradients

Velocity gradients carry the local kinematics: the divergence measures local compression or expansion of the flow, and the vorticity measures local spin. Their two-point correlations tell whether neighbouring regions are compressing together and whether neighbouring haloes spin in the same sense. In this section, we focus on the two-point second order correlations for velocity gradients, namely the divergence field $\theta(\mathrm{x})=\nabla\cdot\mathrm{u}(\mathrm{x})$ and vorticity field $\omega(\mathrm{x})=\nabla\times\mathrm{u}(\mathrm{x})$. The correlation function of velocity divergence can be written as

$$\left\langle\theta\cdot\theta'\right\rangle=\left\langle u_{k,k}\cdot u'_{q,q}\right\rangle=-\delta_{jq}\delta_{kp}Q_{kq,jp}=-Q_{pq,pq}, \tag{89}$$

where the second order derivative of correlation tensor Q (see definition in Eq. (13)) reads

$$\begin{aligned}Q_{pq,pq}=&\frac{\partial Q_{pq}(r)}{\partial r_p\partial r_q}=\frac{\partial^2 A_2}{\partial r^2}r^2+8r\frac{\partial A_2}{\partial r}+12A_2\\&+\frac{\partial^2 B_2}{\partial r^2}+\frac{2}{r}\frac{\partial B_2}{\partial r}=\frac{1}{r^2}\left[\left(r^2L_2\right)_{,r}-2rT_2\right]_{,r}.\end{aligned} \tag{90}$$

The correlation function of the vorticity field can be written as

$$\begin{aligned}\left\langle \omega \cdot \omega^{'} \right\rangle &= \varepsilon_{ijk}\varepsilon_{ipq}\left\langle u_{k,j}u^{'}_{q,p}\right\rangle \\ &= -\left(\delta_{jp}\delta_{kq}-\delta_{jq}\delta_{kp}\right)Q_{kq,jp}.\end{aligned} \tag{91}$$

It can be easily verified that

$$-\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle = \left\langle \omega\cdot\omega^{'}\right\rangle + \left\langle \theta\cdot\theta^{'}\right\rangle. \tag{92}$$

The general expression for $\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle$ is (with the help of Eq. (16))

$$\begin{aligned}\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle &= \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial R_2}{\partial r}\right] \\ &= 3\frac{\partial^2 B_2}{\partial r^2}+\frac{6}{r}\frac{\partial B_2}{\partial r}+r^2\frac{\partial^2 A_2}{\partial r^2}+6r\frac{\partial A_2}{\partial r}+6A_2.\end{aligned} \tag{93}$$

The terms $-\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle$ and $E_u(k)k^2$ form a pair of Fourier transforms

$$-\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle = 2\int_0^\infty E_u(k)k^2\frac{\sin(kr)}{kr}dk. \tag{94}$$

The sum of divergence and vorticity correlation functions can be found in Eqs. (68) and (94)

$$\begin{aligned}R_\theta + R_\omega &= \frac{1}{4r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}S_2^i(r)\right) \\ (R_\theta+R_\omega)(2r) &= \frac{1}{48r^2}\frac{\partial}{\partial r}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^3\frac{\partial}{\partial r}\left(S_2^x(r)r^2\right)\right)\right).\end{aligned} \tag{95}$$

The expression for the vorticity correlation reads (Eq. (91))

$$\begin{aligned}R_\omega = \frac{\left\langle \omega\cdot\omega^{'}\right\rangle}{2} &= -\frac{\partial^2 B_2}{\partial r^2}-\frac{2}{r}\frac{\partial B_2}{\partial r}+3A_2+r\frac{\partial A_2}{\partial r} \\ &= \frac{1}{r^2}\left[r^2\left(A_2 r-\frac{\partial B_2}{\partial r}\right)\right]_{,r}.\end{aligned} \tag{96}$$

The expression for the divergence correlation reads (Eq. (89))

$$\begin{aligned}R_\theta = \frac{\left\langle \theta\cdot\theta^{'}\right\rangle}{2} &= -\frac{1}{2}\left(\frac{\partial^2 B_2}{\partial r^2}+\frac{2}{r}\frac{\partial B_2}{\partial r}+r^2\frac{\partial^2 A_2}{\partial r^2}+8r\frac{\partial A_2}{\partial r}\right. \\ \left.+12A_2\right) &= -\frac{1}{2r^2}\left[r^2\left(4A_2 r+\frac{\partial A_2}{\partial r}r^2+\frac{\partial B_2}{\partial r}\right)\right]_{,r}.\end{aligned} \tag{97}$$

The correlation functions for both divergence and vorticity are fully determined by two scalar functions A_2 and B_2 from the second order velocity correlation tensor Q_{ij} (Eq. (13)) for arbitrary homogeneous and isotropic flow. Both correlation functions can be identified from the N-body simulation (Fig. 28).

3.4.1 *Results for incompressible flow*

The divergence correlation function must be zero everywhere ($\langle\theta\cdot\theta^{'}\rangle = 0$, see Eqs. (97) and (14)). Due to incompressibility, $Q_{ij,ij}=0$, the vorticity correlation function can be reduced to (using Eq. (39))

$$\left\langle \omega\cdot\omega^{'}\right\rangle = -Q_{ii,jj} = 30A_2+18r\frac{\partial A_2}{\partial r}+2r^2\frac{\partial^2 A_2}{\partial r^2}, \tag{98}$$

$$\begin{aligned}\left\langle \omega\cdot\omega^{'}\right\rangle &= -\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle = -\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial R_2}{\partial r}\right] \\ &= -\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial}{\partial r}\left(\frac{1}{r^2}\frac{\partial\left(r^3 L_2\right)}{\partial r}\right)\right],\end{aligned} \tag{99}$$

where the vorticity correlation is fully determined by the longitudinal correlation function $L_2(r)$.

The vorticity correlation can be related to the velocity power spectrum function $E_u(k)$ as

$$R_\omega(r) = \frac{1}{2}\left\langle \omega\cdot\omega^{'}\right\rangle = \int_0^\infty E_u(k)k^2\frac{\sin(kr)}{kr}dk. \tag{100}$$

3.4.2 *Results for constant divergence flow*

All the results for incompressible flow in Section 3.4.1 are still valid for constant divergence flow.

3.4.3 *Results for irrotational flow*

Similarly, the vorticity correlation function should vanish for the irrotational flow ($\left\langle \omega\cdot\omega^{'}\right\rangle = 0$). The divergence correlation function can be reduced to

$$\left\langle \theta\cdot\theta^{'}\right\rangle = -\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle = -\left(r^2\frac{\partial^2 A_2}{\partial r^2}+9r\frac{\partial A_2}{\partial r}+15A_2\right). \tag{101}$$

With Eq. (47), it is related to the transverse correlation T_2 as

$$\begin{aligned}\left\langle \theta\cdot\theta^{'}\right\rangle &= -\nabla^2\left\langle \mathrm{u}\cdot\mathrm{u}^{'}\right\rangle = -\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial R_2}{\partial r}\right] \\ &= -\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial}{\partial r}\left(\frac{1}{r^2}\frac{\partial\left(r^3 T_2\right)}{\partial r}\right)\right].\end{aligned} \tag{102}$$

In addition, it can be related to the velocity power spectrum as

$$R_\theta(r) = \frac{1}{2}\left\langle \theta\cdot\theta^{'}\right\rangle = \int_0^\infty E_u(k)k^2\frac{\sin(kr)}{kr}dk. \tag{103}$$

4 STATISTICAL MEASURES FROM N-BODY SIMULATION

The simulation data described in Section 2 are used to compute all statistical measures described in Section 3. For a given scale r, all particle pairs with a separation r are identified and particle position and velocity data are recorded to compute the correlation and structure functions by averaging that quantity over all pairs with the same separation r, i.e a pairwise averaging for all scales r. This provides a complete view of the variation of these statistical measures on all scales.

4.1 Velocity correlation functions and flow characterization

Figures 3 and 4 present three second order velocity correlation functions that vary with scale r at z = 0. The transverse correlation $T_2(r)$ is positive on all scales, while the longitudinal correlation $L_2(r)$ turns negative at $r \geqslant 21.4Mpc/h$, at which a pair of particles separated by that distance will most likely move in opposite longitudinal directions, but in the same transverse directions. The total velocity correlation $R_2(r)$ becomes slightly negative around r = 75.2 Mpc/h and returns to positive at r=96.6 Mpc/h.

The crossover of $T_2(r)$ and $L_2(r)$ can be found around r_t = 1.3 Mpc/h, a characteristic length scale that divides two different flow regimes. Below r_t, most of the pair of particles is from the same halo. The two-point correlation functions reflect the correlation of particles from the same halo. Longitudinal correlation is dominant over transverse correlation. While above r_t, most pairs are from different

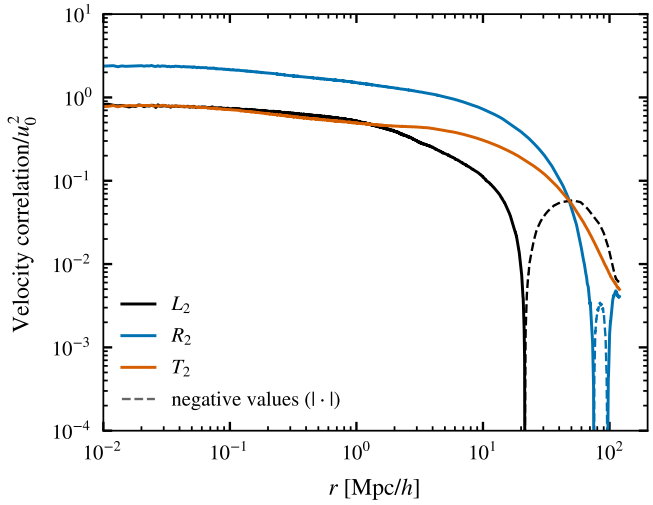


Figure 3. Second-order velocity correlations at $z = 0$, normalized by u_0^2: total R_2, longitudinal L_2 and transverse T_2 (Eqs. 16–18), from all particle pairs in the simulation. Solid lines: positive values; dashed lines: magnitude of negative values. L_2 changes sign at 21.4 Mpc/h; R_2 is negative between 75.2 and 96.6 Mpc/h; T_2 is positive on all scales.

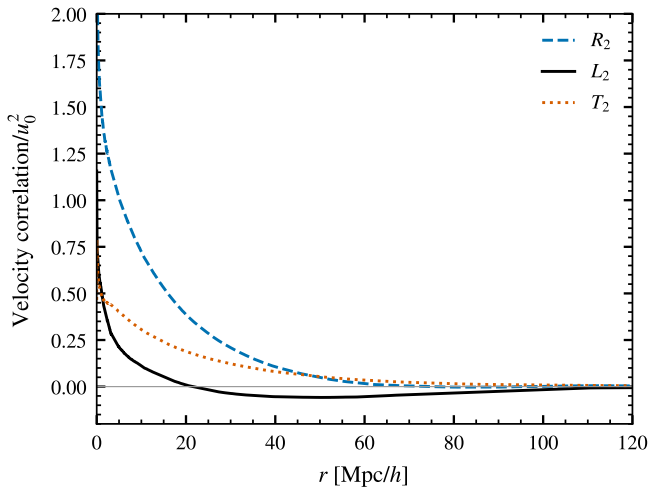


Figure 4. The correlations of Fig. 3 on linear axes ($z = 0$, normalized by u_0^2). Below $r_t \approx 1.3$ Mpc/h most pairs belong to the same halo and $L_2 > T_2$; above r_t most pairs belong to different haloes, the correlations trace halo motions, and $T_2 > L_2$.

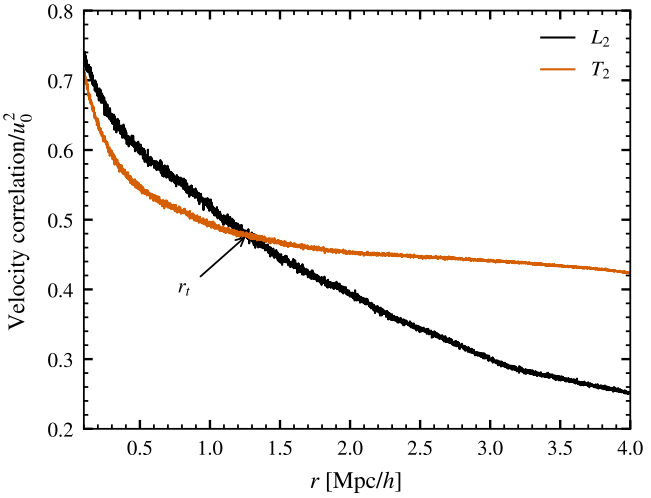


Figure 5. Close-up of the crossover of L_2 and T_2 at $z = 0$ (normalized by u_0^2). The arrow marks $r_t \approx 1.3$ Mpc/h, where the two correlations are equal; it separates the intra-halo regime from the inter-halo regime.

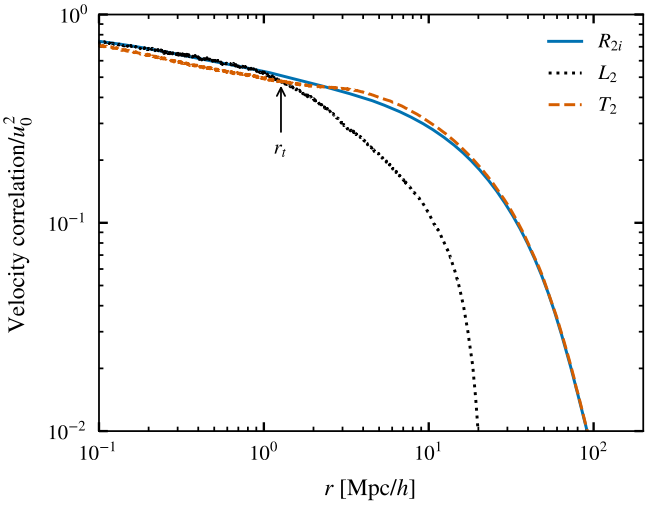


Figure 6. Flow-type test at $z = 0$. The volume-averaged correlation R_{2i} (Eq. 52) equals L_2 for a constant-divergence (or incompressible) flow and T_2 for an irrotational flow. The simulation follows L_2 below r_t (arrow) and T_2 above it: the peculiar velocity is of constant divergence on small scales and irrotational on large scales. Correlations normalized by u_0^2.

haloes. The two-point correlation functions reflect the correlation of halo velocity, and the transverse correlation is dominant.

Obviously, the comoving length scale r_t can be related to the size of the spherical halo with the characteristic mass $m_h^*(a)$. With $m_h^*(a) \propto a^{3/2}$ (see reference [42]) and constant density ratio Δ_c, we expect $r_t(a) \propto \left(m_h^*/(\Delta_c \rho_0)\right)^{1/3} \propto a^{1/2}$ (Fig. 11).

Next, the kinematic relations developed in Section 3.2 (Eq. (52)) can be applied to identify the nature of flow on small and large scales. Figure 6 plots the variation of $R_{2i}(r)$ with scale r, along with two correlation functions $L_2(r)$ and $T_2(r)$. In Fig. 6, on small scales $r \ll r_t$, $R_{2i} \approx L_2$ means a constant divergence flow for peculiar velocity on small scales ($\nabla \cdot \mathrm{u} = const$). While on large scales $r \gg r_t$, $R_{2i} \approx T_2$ means an irrotational velocity field ($\nabla \times \mathrm{u} = 0$). The critical length scale $r_t \approx 1.3 Mpc/h$ is defined as the scale at which $L_2(r) = T_2(r)$ (indicated by an arrow). The irrotational flow on large scales is expected, where the Zeldovich approximation and the linear perturbation theory apply. Unlike incompressible flow that is incompressible on all scales, the peculiar velocity field of dark matter flow has a scale-dependent behavior, i.e. a constant divergence flow on small scale and an irrotational flow on large scale.

Figure 7 presents the transverse velocity correlation function $T_2(r)$ at different redshifts z. Correlation functions are normalized by $u^2(a)$, that is, the dispersion of the one-dimensional velocity at different redshifts (Eq. (1)). Figure 7 hints at an exponential form for $T_2(r)$ on large scales at different z. The detailed models for the

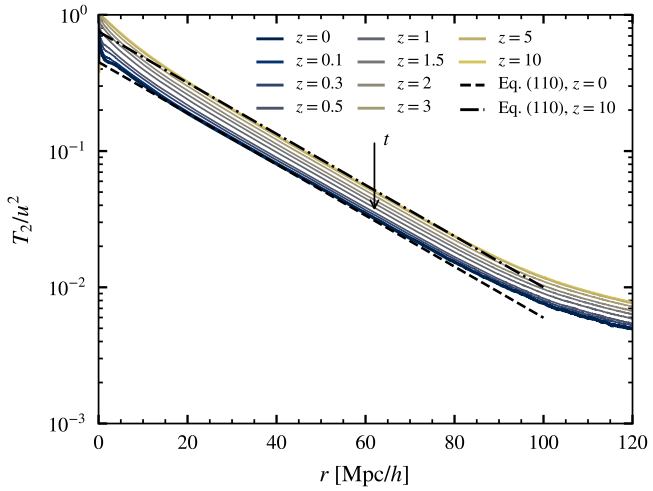


Figure 7. Transverse correlation $T_2(r)/u^2(a)$ on log-linear axes for $z = 0$–10 (colour; $z = 0$ darkest). Straight lines on these axes indicate the exponential form of Eq. (110); black lines are that model for $z = 0$ (dashed) and $z = 10$ (dash-dotted) with $r_2 = 23.13\,\mathrm{Mpc}/h$. The arrow indicates increasing time.

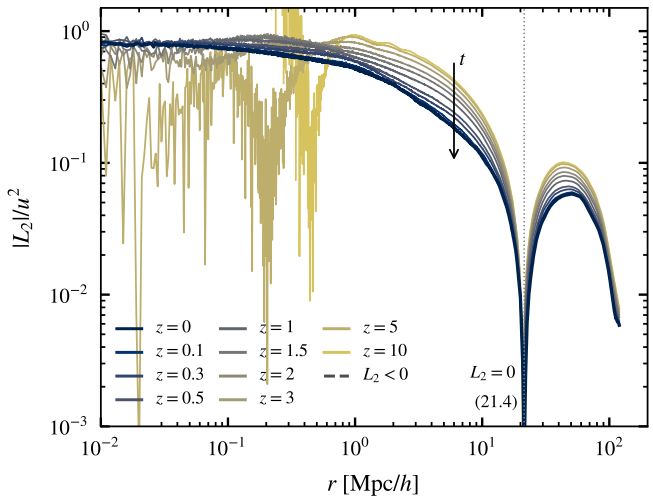


Figure 8. Longitudinal correlation $L_2(r)/u^2(a)$ for $z = 0$–10 (solid: positive; dashed: magnitude where negative). Correlation on small scales develops after $z \approx 3$ as haloes form; the noisy curves at $r < 1\,\mathrm{Mpc}/h$ for $z \geqslant 5$ reflect the few close pairs at early times. The zero crossing of L_2, at $r \approx 21.4\,\mathrm{Mpc}/h$, does not move with time; in the large-scale model of Section 5.1 it occurs at $r = r_2$. Black lines: Eq. (111) for $z = 0$ and $z = 10$ with $r_2 = 23.14\,\mathrm{Mpc}/h$ (dashed) and $r_2 = 21.4\,\mathrm{Mpc}/h$ (dotted).

correlation functions are provided in Section 5 (Eqs. (110) and (140) for large and small scales, respectively).

Figure 8 presents the longitudinal velocity correlation $L_2\,(r)$ (normalized by $u^2\,(a)$). With the formation of more and more haloes, the number of pairs of particles from the same halo with small separation r increases with time, along with the gradual development of longitudinal correlation on small scales after z=3. The longitudinal correlation becomes negative beyond a fixed comoving separation, $r \approx 21.4\,\mathrm{Mpc}/h$, that does not change with time; in the model of Section 5.1 this zero is at $r = r_2$. The length scale r_2 is related to the horizon size at the matter-radiation equality (see Section 5.1.3). The

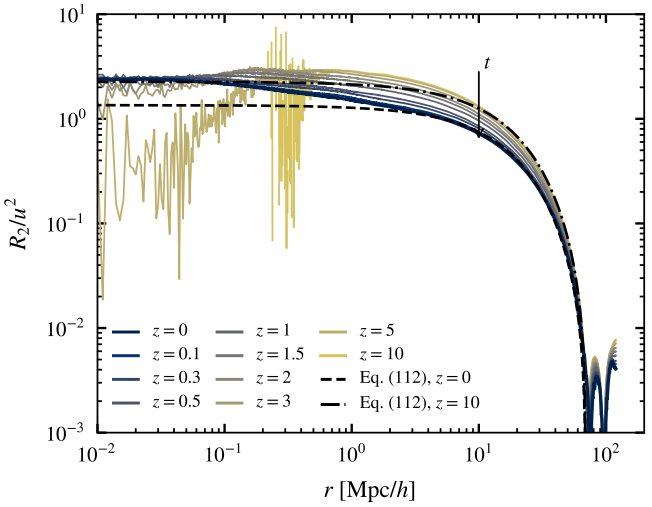


Figure 9. Total correlation $R_2(r)/u^2(a)$ (Eq. 23) for $z = 0$–10, with the large-scale model Eq. (112) for $z = 0$ (dashed) and $z = 10$ (dash-dotted). Dashed coloured segments show the magnitude of negative values.

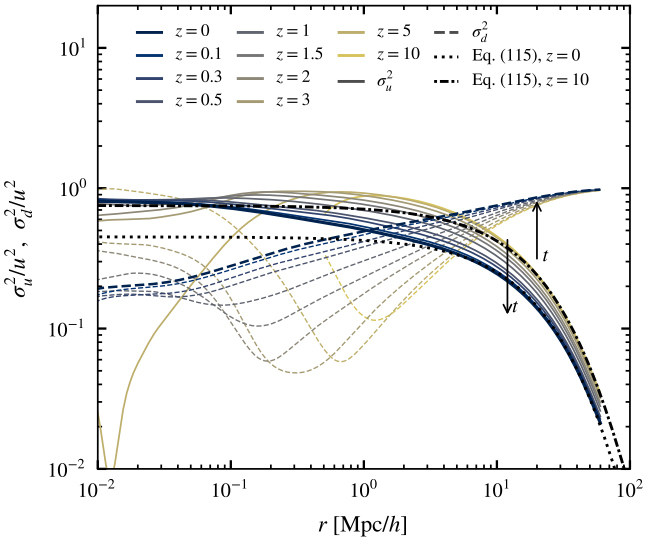


Figure 10. Dispersion functions normalized by $u^2(a)$ for $z = 0$–10: $\sigma_u^2(r)$, the energy in scales larger than r (solid; from R_2 via Eq. 30), and $\sigma_d^2(r)$, the energy in scales smaller than r (dashed; Eq. 31). Black dotted lines: large-scale model Eq. (115) for $z = 0$ and $z = 10$. The growth of σ_d^2 with time shows energy accumulating on small scales as haloes form.

models are presented in Section 5 (Eqs. (111) and (138) for large and small scales, respectively).

Figure 9 presents the total velocity correlation function $R_2\,(r)$ at different redshift z (from Eq. (23)). The detail models are presented in Section 5 (Eqs. (112) and (139)) for large and small scales.

4.2 Velocity dispersion functions and energy distribution

Figure 10 presents the variation of two dispersion functions with comoving scale r for different redshift z. The dispersion of smoothed velocity $\sigma_u^2\,(r)$ is obtained from Eq. (30) (solid line), inverted by three cumulative trapezoidal integrations from $r = 0$ on the $1\,\mathrm{kpc}/h$

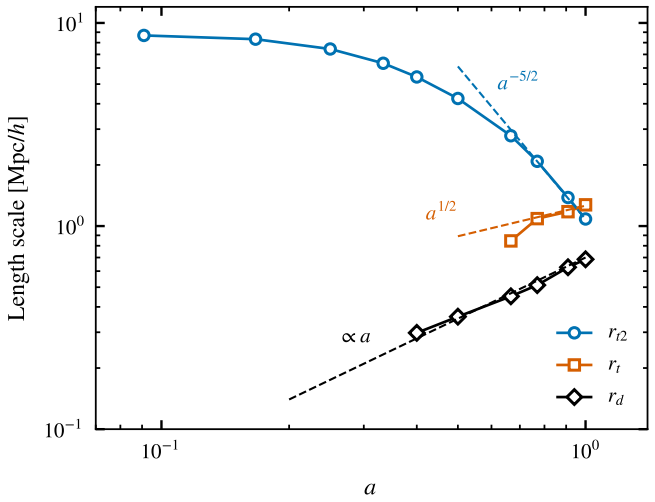


Figure 11. Characteristic comoving scales versus scale factor a: r_{t2}, where $\sigma_u^2 = \sigma_d^2$ (half of the kinetic energy lies below r_{t2}); r_t, where $L_2 = T_2$; and r_d, where $\langle u_L^2 \rangle$ peaks (Eq. 136, Fig. 24). Dashed lines show the scalings quoted in the text, $r_{t2} \propto a^{-5/2}$, $r_t \propto a^{1/2}$ and $r_d \propto a$. The r_{t2} points follow $a^{-5/2}$ for $a \gtrsim 0.67$; r_t is resolved (a unique crossing) at the four latest outputs only. r_t and r_d are located after a 0.1 Mpc/h running mean.

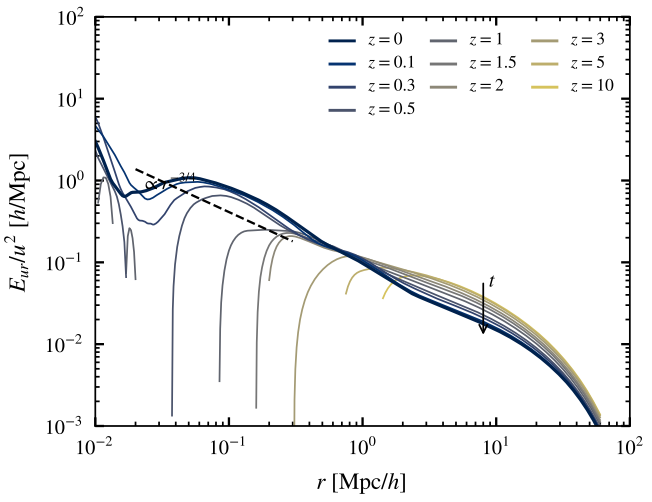


Figure 12. Real-space energy distribution $E_{ur}(r) = -\partial\sigma_u^2/\partial r$ (Eq. 32), normalized by $u^2(a)$ (units $h\,\mathrm{Mpc}^{-1}$), for $z = 0$–10. With time the energy on large scales decreases and the energy on small scales increases. The dashed segment shows the slope $r^{n-1} = r^{-3/4}$ implied by $\sigma_d^2 \propto r^{1/4}$ (Eq. 141); below 0.1 Mpc/h the measured E_{ur} is noisy (least-squares slope -0.9 over 0.05–0.5 Mpc/h).

grid; with this rule σ_d^2 follows the $n = 1/4$ law of Eq. 141, e.g. 0.23, 0.28 and 0.34 at $r = 0.05$, 0.1 and 0.2 Mpc/h against 0.24, 0.29 and 0.34 from Eq. 141, while the velocity dispersion $\sigma_d^2(r)$ is obtained from Eq. (31) (dash line). The increasing small-scale power with time can be seen from the increasing $\sigma_d^2(r)$ for a given r. The dispersion function $\sigma_u^2(r)$ on large scales can be modeled by Eq. (115) and plotted on a dotted line in good agreement with the N-body simulation (solid blue line). Setting the energy on the scales above r equals the energy contained in scales smaller than r, i.e. $\sigma_u^2(r_{t2}) = \sigma_d^2(r_{t2}) = 0.5$, a second comoving length scale r_{t2} can be introduced. The kinetic energy contained in scales less than r_{t2} is 50% of the total kinetic energy. Figure 11 presents the variation of $r_{t2} \propto a^{-5/2}$ and $r_t \propto a^{1/2}$ (the r_{t2} scaling describes the late-time points, $a \gtrsim 0.67$; r_t is resolved only at the four latest outputs). The length scale r_{t2} that decreases with time reflects the increasing fraction of kinetic energy contained in small scales with more haloes being formed.

Figure 12 presents the energy distribution in real space from Eq. (32). Kinetic energy contained in large scale is decreasing with time, with more energy contained in small scale. With more haloes forming, the small scale power ($\propto u^2 \propto a^{3/2}$ Eq. (139)) increases faster than the large scale ($\propto a_0 u^2 \propto a^1$, Eq. (112)) with increasing fraction of kinetic energy contained in small scales.

4.3 Velocity structure function S_2^x and enstrophy distribution

Figure 13 presents the variation of structure function $S_2^x(r)$ (from Eq. (76)) with comoving scale r for different redshifts z (solid lines). The corresponding models of $S_2^x(r)$ on large and small scales are discussed in Section 5 (Eqs. (117) and (143)). The one-dimensional enstrophy of smoothed velocity $S_2^x(r)/(2r^2)$ (Eq. (73), normalized by u^2 with unit of $(h/Mpc)^2$), i.e., the enstrophy contained in all scales above r, is also plotted in dashed lines. The measured S_2^x agrees with the small-scale model Eq. (143) at 0.1–0.3 Mpc/h (e.g. $0.211u^2$ against $0.214u^2$ at 0.3 Mpc/h).

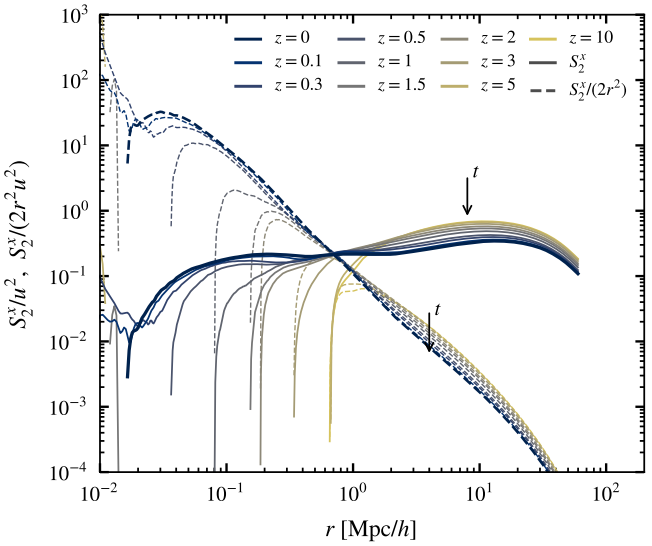


Figure 13. Enstrophy structure function $S_2^x(r)/u^2$ (solid; Eq. 72) and the enstrophy contained in scales larger than r, $S_2^x/(2r^2u^2)$ (dashed; Eq. 73, units $h^2\,\mathrm{Mpc}^{-2}$), for $z = 0$–10. Arrows indicate increasing time.

Figure 14 presents the distribution of enstrophy $E_{nr}(r)$ (Eq. (73)) in real space r at different redshifts z. With more haloes formed with finite angular velocity (or vorticity), the enstrophy distribution gradually extends to smaller and smaller scales.

4.4 Velocity structure functions S_2^l and one-fourth law

Figure 15 presents the variation of different velocity dispersion and structure functions with scale r at z=0. From the N-body simulation, we can directly obtain the velocity dispersion $\sigma_d^2(r)$ (Eq. (29)), dispersion of smoothed velocity $\sigma_u^2(r)$ (Eq. (27)), total velocity struc-

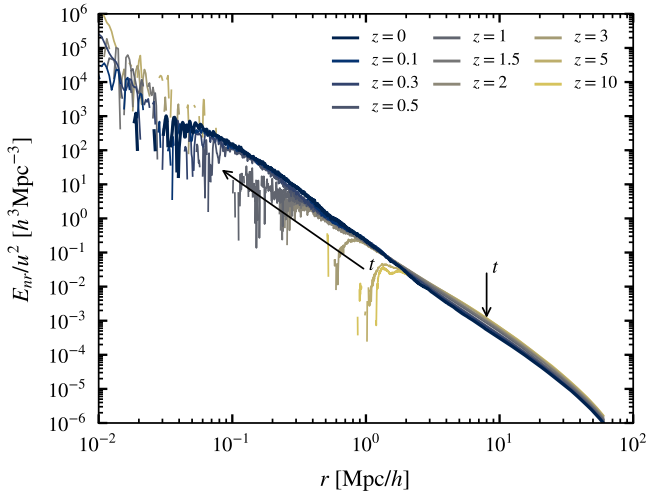


Figure 14. Real-space enstrophy distribution $E_{nr}(r)$ (Eq. 78), normalized by $u^2(a)$ (units $h^3\,\mathrm{Mpc}^{-3}$), for $z=0$–10. In comoving coordinates the enstrophy on small scales increases with time and that on large scales decreases, as spinning haloes form on ever smaller scales.

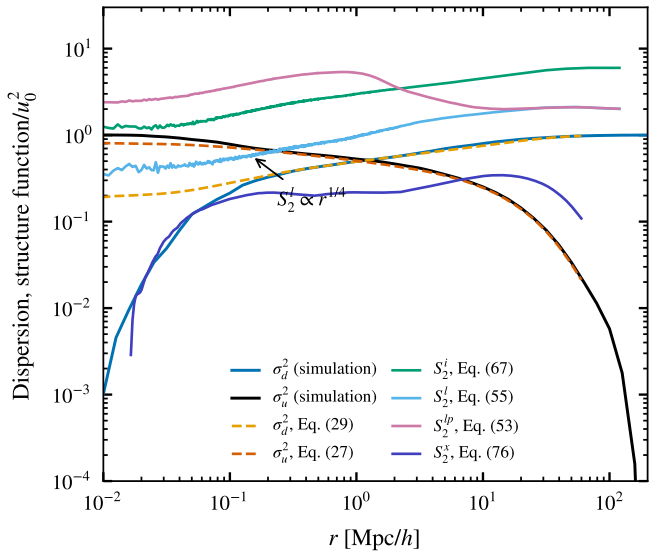


Figure 15. Dispersion and structure functions at $z=0$, normalized by u_0^2. Measured directly from particle pairs: σ_d^2 and σ_u^2 (Eqs. 29, 27), S_2^i (Eq. 67), S_2^l (Eq. 55) and S_2^{lp} (Eq. 53). Obtained indirectly from R_2 for comparison: σ_u^2 and σ_d^2 (Eqs. 30, 31) and S_2^x (Eq. 76). Legend equation numbers refer to the definitions used. On small scales $S_2^l \propto \sigma_d^2 \propto r^{1/4}$; the two longitudinal structure functions differ ($S_2^{lp} \neq S_2^l$), a signature of collisionless flow (Eqs. 59–61).

ture function $S_2^i(r)$ (Eq. (67)), longitudinal structure functions $S_2^l(r)$ and $S_2^{lp}(r)$ (Eqs. (53) and (55)), and the structure function $S_2^x(r)$ from Eq. (76).

The dispersion $\sigma_u^2(r)$ and $\sigma_d^2(r)$ can also be obtained indirectly using the correlation function $R_2(r)$ (Eqs. (30) and (31)), which show good agreement on large scales with those directly obtained from simulation. On small scales, the velocity dispersion $\sigma_u^2(r)$ obtained indirectly from R_2 (Eq. (30)) still matches the one directly obtained from the N-body simulation, while the velocity dispersion $\sigma_d^2(r)$ obtained through $\sigma_d^2(r) = u^2 - \sigma_u^2(r)$ has some disagreement with that directly obtained from the simulation. The potential reasons can be: 1) The sum of two dispersions $\sigma_d^2(r) + \sigma_u^2(r)$ does not exactly equal u^2 in N-body simulation. This might be related to the simulation resolution as a result of finite gravitational softening. 2) The numerical artifacts introduced from the grid might have an effect on the statistical isotropy on small scales. Since Eq. (30) is derived on the basis of the homogeneity and isotropy, violation of local statistical isotropy may also contribute to this discrepancy.

A power law of $S_2^l(r) \propto \sigma_d^2(r) \propto r^{1/4}$ can be identified on small scales. The dispersion function $\sigma_d^2(r)$ should share the same power law scaling as structure functions $S_2^i(r)$ (Eq. (70)) and $S_2^l(r)$ (Eq. (82)). The reason for a power-law scaling is briefly outlined here. Let us first compute the potential energy for a sphere of radius r

$$U(r) = -\frac{\int_0^r \frac{G}{y} M(y)\,\rho(y)\,4\pi y^2 dy}{\int_0^r \rho(x)\,4\pi x^2 dx}, \tag{104}$$

where $M(y) = \int_0^y \rho(x)\,4\pi x^2 dx$ is the total mass in all shells with a radius smaller than y.

The density $\rho(y)$ in that sphere can be related to the correlation function $\xi(r)$. By assuming a power-law of $\xi(r)$ ($\xi(r) = (r/6.1)^{-1.75}$ in reference [43]), density $\rho(y)$ reads

$$\begin{aligned} \xi(r) &= a^m \left(r/r_\xi\right)^{-n} \\ \rho(y) &= \rho_0\,(1+\xi(y)) \approx \rho_0 a^m \left(y/r_\xi\right)^{-n}, \end{aligned} \tag{105}$$

where $r_\xi \approx 3.08 Mpc/h$, $m \approx 2$, $n \approx 1.75$, and ρ_0 is the mean density at z=0. Inserting density into Eq. (104),

$$\begin{aligned} U(r) &= -\frac{3-n}{5-2n}\frac{GM(r)}{r} \\ &= -\frac{4\pi G\rho_0 a^m r^2}{5-2n}\left(\frac{r}{r_\xi}\right)^{-n} = -\frac{3a^m H_0^2 r^2}{2(5-2n)}\left(\frac{r}{r_\xi}\right)^{-n}. \end{aligned} \tag{106}$$

The specific kinetic energy of that sphere can be written as $T(r) = 3/2\sigma_d^2$, where σ_d^2 is the one-dimensional velocity dispersion of particles in that sphere. Applying the virial theorem $2T(r) + \gamma U(r) = 0$ leads to the equation (where γ is the virial ratio),

$$\sigma_d^2(r) = \frac{\gamma (H_0 r)^2}{2r^5}\frac{\int_0^r (1+\xi(y))\left(1+\bar{\xi}(y)\right) y^4 dy}{\left(1+\bar{\xi}(r)\right)}. \tag{107}$$

For power-law density correlation in Eq. (105)

$$\sigma_d^2(r) = \frac{\gamma a^m (H_0 r)^2}{2(5-2n)}\left(\frac{r}{r_\xi}\right)^{-n}. \tag{108}$$

The structure function can be computed from Eq. (82) (peculiar velocity is of constant divergence on small scales, where Eq. (82) can be applied) as

$$S_2^l(r) = \frac{\gamma(6-n)(8-n)}{24(5-2n)\,2^{2-n}} a^m (H_0 r)^2 \left(\frac{r}{r_\xi}\right)^{-n}. \tag{109}$$

Please note that the constant divergence flow shares the same kinematic relations as that of incompressible flow on second order (see Section 3.3.6). With $n = 7/4$, we can estimate $\sigma_d^2(r) \approx 0.19\gamma a^m r^{1/4} u_0^2$ and $S_2^l(r) \approx 0.3533\gamma a^m r^{1/4} u_0^2$, both of which follow a power law scaling $\propto r^{1/4}$. The detailed models for structure functions are provided in Section 5 (Eqs. (116) and (137)).

Finally, the velocity dispersion tensor $\sigma_{dij}^2(r)$ was found from simulation by computing the average velocity dispersion within a sphere of radius r centered around randomly selected particles. Figure 16 plots the six components of the velocity dispersion tensor

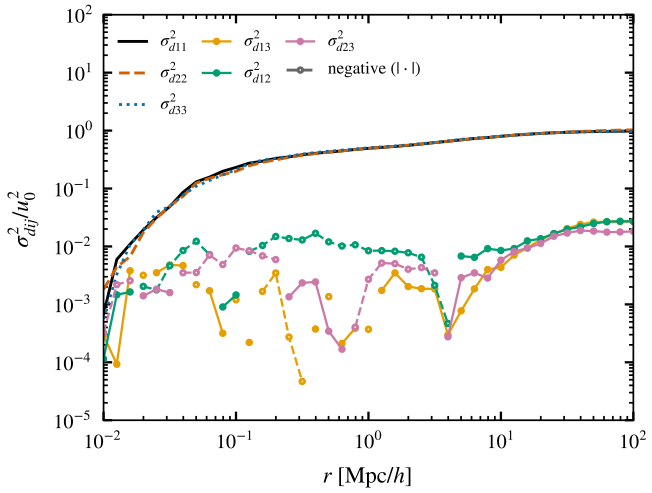


Figure 16. The six components of the velocity dispersion tensor $\sigma^2_{dij}(r)$ at $z = 0$, normalized by u_0^2, from spheres of radius r centred on randomly selected particles. Off-diagonal components (with markers) are shown as solid lines where positive and dashed where negative. The diagonal components are equal on all scales and the off-diagonal ones are one to two orders of magnitude smaller: the dispersion tensor is isotropic.

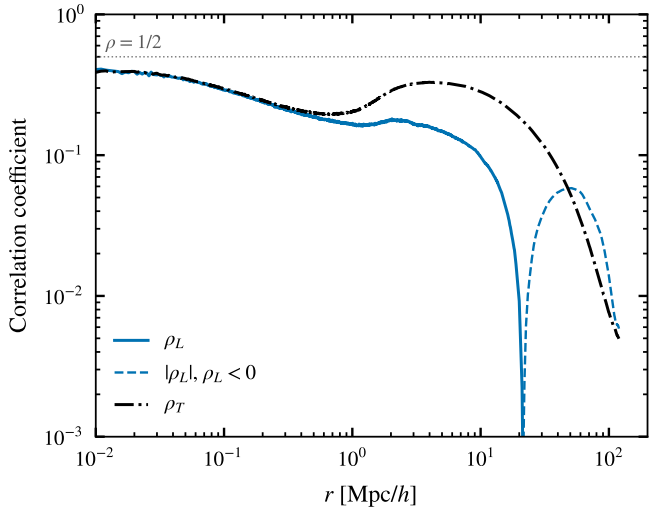


Figure 17. Velocity correlation coefficients ρ_L and ρ_T (Eq. 20) at $z = 0$. $\rho_T > 0$ on all scales; ρ_L becomes negative for $r \gtrsim 21.4$ Mpc/h (dashed: magnitude). Longitudinal and transverse velocities are strongly correlated on small scales, but much less than in incompressible collisional flow, where $\rho_L = \rho_T = 1$; the dotted line marks the estimate $\rho_L(0) = 1/2$ of Section 3.3.1. At the smallest separations resolved (0.001–0.03 Mpc/h) both coefficients are ≈ 0.39.

varying with scale r, where red color indicates that the value is negative. The off-diagonal terms are negligible. The velocity dispersion tensor is isotropic with diagonal terms $\sigma^2_{d11} \approx \sigma^2_{d22} \approx \sigma^2_{d33}$ on all scales.

4.5 Longitudinal and transverse velocities on different scales

Figure 17 presents the correlation coefficients of longitudinal and transverse velocities ($\rho_L(r)$ and $\rho_T(r)$ in Eq. (20)) varying with scale r. The transverse correlation is positive $\rho_T(r) > 0$ for all

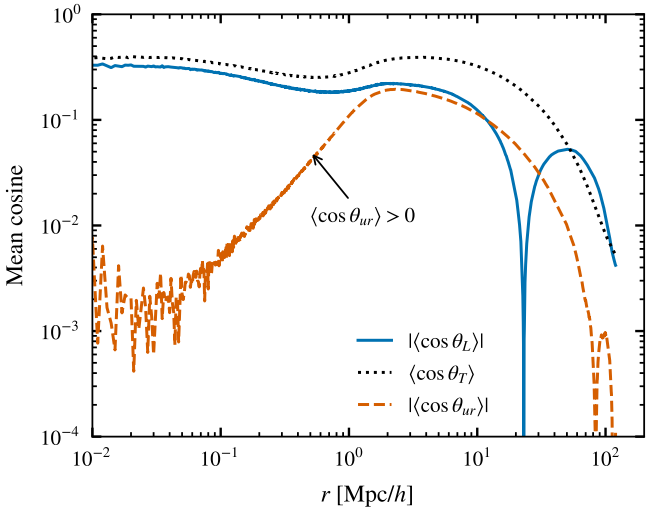


Figure 18. Mean cosines of the angles between longitudinal velocities (θ_L), transverse velocities (θ_T), and velocity and separation vector (θ_{ur}), Eqs. (21) and (22), at $z = 0$; magnitudes are plotted; $\langle\cos\theta_{ur}\rangle$ is positive (arrow) for **r** pointing to the partner (Eq. 22). Unlike in incompressible turbulence, **u** and **r** are not statistically independent, because $\langle u_L\rangle = \langle u_i\hat{r}_i\rangle \neq 0$; they are uncorrelated only on very small and very large scales, where $\langle u_L\rangle \to 0$.

scales, while the longitudinal correlation becomes negative for $r \gtrsim 21.4$ Mpc/h. It can be shown that u_L (or u_T) and $u_L^{'}$ (or $u_T^{'}$) are strongly correlated (both u_L and $u_L^{'}$ point in the same direction) on small scales with both ρ_L and ρ_T approaching 0.5 for small r. This is a distinct feature of SG-CFD (Eq. (58)) discussed in Section 3.3.1. On the contrary, the velocity is fully correlated $\rho_L = \rho_T = 1$ for incompressible collisional flow. On large scales with $r > r_2$, the longitudinal velocities u_L and $u_L^{'}$ are inversely correlated. The increase in the correlation coefficient between scales 1 and 4 Mpc/h may be due to the sharp drop in velocity dispersion $\left\langle |\mathrm{u}_T|^2 \right\rangle$ and $\langle u_L^2\rangle$ (Fig. 24).

Similarly, Fig. 18 plots the variation of $\langle\cos\theta_L\rangle$, $\langle\cos\theta_T\rangle$ and $\langle\cos\theta_{ur}\rangle$ (defined in Eqs. (21) and (22)) with scale r at z=0. Note that the velocity **u** and vector **r** fields are independent in incompressible homogeneous turbulence. This is not true for SG-CFD due to $\langle u_L\rangle = \langle \mathrm{u}\cdot\mathrm{r}\rangle \neq 0$. Both fields are only statistically uncorrelated on small or large scales, but are correlated on intermediate scales.

To further investigate the statistical properties of the fields **u** and **r**, we identify all pairs of particles with a given separation r and compute the histogram of three angles θ_{ur}, θ_T, and $\theta_{\mathrm{uu'}}$ in Eqs. (21) and (22). Figures 19 and 20 present the probability distribution of the cosine of three angles (i.e. $\cos\theta_{\mathrm{ur}}$, $\cos\theta_T$ and $\cos\theta_{\mathrm{uu'}}$) and the distribution of angles (i.e. θ_{ur}, θ_T, and $\theta_{\mathrm{uu'}}$) on three different scales of r at z=0.

The cosine of angle between **u** and **r,** i.e. $\cos\theta_{\mathrm{ur}}$, has a uniform distribution on both small and large scales, where all pairs are either from the same or from different haloes (r=0.1Mpc/h and r=100Mpc/h). Namely, **u** and **r** fields are uncorrelated on small and large scales. For a particle with a given **u**, the unit vector $\hat{\mathrm{r}}$ is uniformly distributed on a unit sphere (distribution is isotropic with the same probability of finding another particle separated by r along any direction). However, on intermediate scales where pairs of particles can be either from same haloes or different haloes ($r_t = 1.3Mpc/h$), the skewed distribution of angle θ_{ur} indicates that for a particle with given **u**, probability to find other particles is not isotropic and prefers

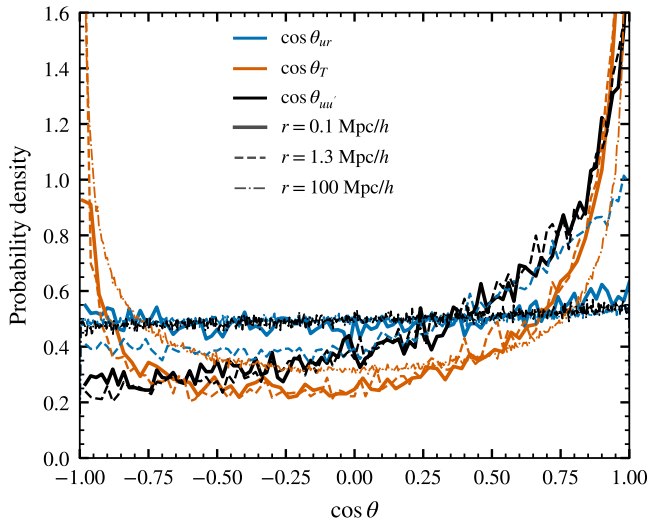


Figure 19. Probability densities of $\cos\theta_{ur}$, $\cos\theta_T$ and $\cos\theta_{uu'}$ at $z = 0$ for separations $r = 0.1$ (intra-halo), 1.3 ($\approx r_t$) and 100 Mpc/h (large scale). For each pair θ_{ur} is measured for both particles, each with respect to the unit vector pointing to its partner. A flat density (value 1/2) means no preferred direction. $\cos\theta_{ur}$ is uniform at small and large r but skewed towards positive values at r_t, where **u** and **r** are correlated.

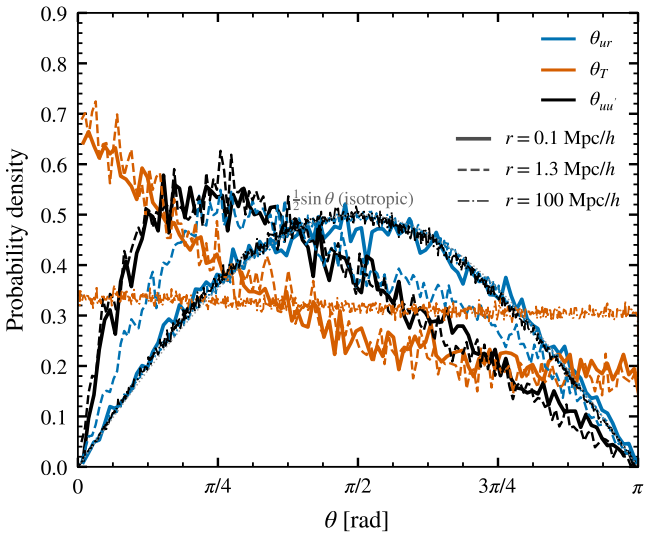


Figure 20. Probability densities of the angles θ_{ur}, θ_T and $\theta_{uu'}$ for the separations of Fig. 19 ($z = 0$); the grey dotted curve, $\frac{1}{2}\sin\theta$, is the isotropic distribution. At intermediate r the skew of θ_{ur} towards $\theta < \pi/2$ shows that **u** and **r** are correlated. θ_T and $\theta_{uu'}$ favour angles below $\pi/2$ on small scales, where gravity correlates the two velocities, and become uniform in θ_T (and in $\cos\theta_{uu'}$) on large scales.

an angle $\theta_{\rm ur} < \pi/2$: with $\mathbf{r} = \mathbf{x}' - \mathbf{x}$ pointing to the partner, as defined in Eq. (22), a particle tends to move towards the other particles, the signature of the mean infall $\langle \Delta u_L \rangle < 0$.

The distribution of angle θ_T skews toward $\theta_T < \pi/2$ on small scales where two transverse velocities are strongly correlated. The angle θ_T changes to a uniform distribution between [0 π] on large scales where two transverse velocities u_T and u_T' are not correlated.

The angle $\theta_{\rm uu'}$ also skews toward $\theta_{uu'} < \pi/2$ on small scales due to strong gravity and $\cos\theta_{\rm uu'}$ has a uniform distribution on large scale where u and u$'$ are not correlated. On large scales, the possibility to find the velocity u$'$ is the same along any direction for a given velocity **u**.

5 MODELS FOR VARIOUS STATISTICAL MEASURES

In this section, we present models for second order statistical measures from N-body simulation. The kinematic relations in Section 3 and the simulation results in Section 4 can be combined to provide simple models for density, velocity, and potential fields. Since the self-gravitating collisionless dark matter flow has two distinct regimes: 1) irrotational flow on large scales, and 2) constant divergence flow on small scale, models for each regime are discussed separately followed by combined models for the entire range.

5.1 Statistical measures on large scale (linear regime)

5.1.1 Velocity correlation/dispersion/structure functions

The N-body simulation (Fig. 7) strongly suggests an exponential form for transverse velocity correlation function on large scale

$$T_2(r,a) = a_0 u^2 \exp\left(-\frac{r}{r_2}\right), \tag{110}$$

which should be an intrinsic property of large-scale dynamics. Here, the amplitude $a_0(a)$ is a function of time and $r_2 = 23.13$ Mpc/h is a characteristic length scale (best fit at $z = 0$, Table 2; the model curves of Figs. 7, 9 and 10 use $r_2 = 23.13$–23.14 Mpc/h, and Fig. 8 also shows the model with $r_2 = 21.4$ Mpc/h, the measured zero of L_2). This exponential form is not a coincidence and must be deeply rooted in dynamics and kinematics on large scales, which is demonstrated in a separate paper on the dynamic relations between correlation functions of different order [29].

In the present context, what is important is that this provides a very simple and good description of velocity correlation on large scales. Figure 21 plots the variation of prefactor $a_0(a)$ (obtained by fitting Eq. (110) to data in Fig. 7 by least squares over $20 < r < 90$ Mpc/h with $r_2 = 23.1321$ Mpc/h fixed) and velocity dispersion u^2 (in Eq. (1)) with scale factor a. Clearly, $a_0 u^2 \propto a$ such that $a_0 (u/u_0)^2 = 0.45a$, where $u_0^2 = u^2(a = 1)$ is the dispersion in the present epoch. On large scales, the kinetic energy increases $\propto a_0 u^2 \propto a$.

With kinematic relations developed in Section 3, other correlation functions of the same order can be derived easily. For irrotational flow, the relation between three scalar correlation functions (Eq. (47)) enables us to derive the longitudinal correlation function,

$$L_2(r,a) = a_0 u^2 \exp\left(-\frac{r}{r_2}\right)\left(1 - \frac{r}{r_2}\right) \tag{111}$$

with $L_2 = 0$ at $r = r_2$. The plot of L_2 at different redshifts z in Fig. 8 clearly indicates that r_2 should be relatively independent of time or redshift; the measured zero of L_2 lies between 21.0 and 21.4 Mpc/h at all redshifts (21.4 Mpc/h at $z = 0$), 7–9 per cent below the fitted r_2.

The total correlation function (see Fig. 9) reads

$$R_2(r,a) = \left\langle \mathrm{u} \cdot \mathrm{u}' \right\rangle = 2R(r) = a_0 u^2 \exp\left(-\frac{r}{r_2}\right)\left(3 - \frac{r}{r_2}\right), \tag{112}$$

with $R_2 = 0$ at $r = 3r_2$. The mth order moment of $R_2(r)$ is

$$\int_0^\infty R_2(r)\, r^m dr = (2-m)\Gamma(1+m)\, r_2^{1+m} a_0 u^2, \tag{113}$$

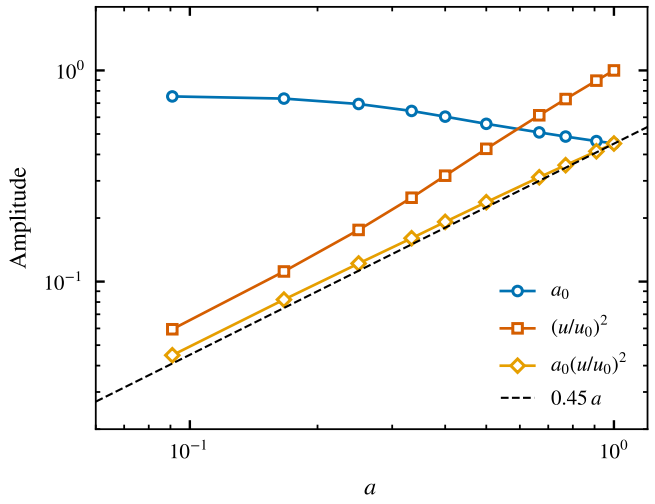


Figure 21. Amplitude $a_0(a)$ of the exponential transverse correlation (Eq. 110; fitted to the data of Fig. 7), the squared rms velocity $(u/u_0)^2$, and their product, versus scale factor a. The product grows linearly, $a_0(u/u_0)^2 = 0.45\,a$ (dashed), so the large-scale kinetic energy grows as a. $u_0^2 = u^2(a=1)$.

where $a_0u^2 = 0.45au_0^2$ and $r_2 = 23.13Mpc/h$. In principle, the constant comoving length scale r_2 should be related to the horizon size at the matter-radiation equality, as shown in Eq. (134).

The correlation length of the velocity field (Eq. (51)),

$$l_{u0} = \frac{1}{u^2}\int_0^\infty R(r)\,dr = \frac{1}{2u^2}\int_0^\infty R_2(r)\,dr = a_0 r_2. \tag{114}$$

The velocity dispersion $\sigma_u^2(r)$ on scale r reads (from Eq. (30))

$$\sigma_u^2(r) = \frac{3}{2}a_0u^2\left(\frac{r_2}{r}\right)^2\left\{-15\left(\frac{r_2}{r}\right)^4+3\left(\frac{r_2}{r}\right)^2+\exp\left(-\frac{2r}{r_2}\right)\cdot\left[15\left(\frac{r_2}{r}\right)^4+30\left(\frac{r_2}{r}\right)^3+27\left(\frac{r_2}{r}\right)^2+14\left(\frac{r_2}{r}\right)+4\right]\right\}, \tag{115}$$

where $\sigma_u^2(r) \propto r^{-4}$ for $r\to\infty$ and $\sigma_u^2(r) = a_0u^2 \propto a$ for $r\to 0$. Comparison with the N-body simulation is shown in Fig. 10.

The second order longitudinal structure function $S_2^l(r)$ on large scale reads (from Eq. (88))

$$S_2^l(r) = 2u^2\left[1+a_0\left(\frac{r}{r_2}-1\right)\exp\left(-\frac{r}{r_2}\right)\right], \tag{116}$$

while the second order structure function $S_2^x(r)$ can be found from Eqs. (67) and (76)

$$S_2^x(r) = \frac{3}{2}a_0u^2\left\{\left[1+3\left(\frac{r_2}{r}\right)^2\right]\left(\frac{r_2}{r}\right)^2 - \exp\left(\frac{-2r}{r_2}\right)\left[1+\left(\frac{r_2}{r}\right)^2\right]\left[3\left(\frac{r_2}{r}\right)^2+6\left(\frac{r_2}{r}\right)+4\right]\right\}. \tag{117}$$

The correlation function of the vorticity field vanishes on large scales $R_\omega = 0$ due to the irrotational nature, whereas the correlation of divergence field can be derived from Eq. (102)

$$R_\theta(r,a) = \frac{1}{2}\left\langle(\nabla\cdot\mathrm{u})\left(\nabla\cdot\mathrm{u}'\right)\right\rangle = \frac{a_0u^2}{2rr_2}\exp\left(-\frac{r}{r_2}\right)\left[\left(\frac{r}{r_2}\right)^2-7\left(\frac{r}{r_2}\right)+8\right], \tag{118}$$

where vanishing divergence correlation $R_\theta = 0$ at two locations of $\left(7\pm\sqrt{17}\right)r_2/2$ (33.3 and 128.7Mpc/h).

5.1.2 *Density/potential correlation functions*

We first introduce a log-density field $\eta(x) = \log(1+\delta) \approx \delta$ and $\eta(x) \approx \delta(x)$ for small overdensity δ. For weakly nonlinear regime, the well-established Zeldovich approximation [39] and the linear perturbation theory relate the density $\delta(\mathrm{x})$ to potential $\phi(\mathrm{x})$ and velocity fields,

$$\delta \approx \eta = -\frac{\nabla\cdot\mathrm{u}}{aHf(\Omega_m)}, \quad \mathrm{u} = -\frac{Hf(\Omega_m)\nabla\phi}{4\pi G\rho a}, \quad \text{and} \quad \delta \approx \eta = \frac{\nabla^2\phi}{4\pi G\rho a^2}, \tag{119}$$

where $f(\Omega_m) \approx \Omega_m^{0.6}$ is a function of matter content Ω_m and $f(\Omega_m = 1) = 1$ for matter dominant model. With the help of Eqs. (102) and (119), the density correlation on large scale should read

$$\xi(r,a) = \left\langle\delta(\mathrm{x})\cdot\delta\left(\mathrm{x}'\right)\right\rangle = \frac{\left\langle\theta(\mathrm{x})\cdot\theta\left(\mathrm{x}'\right)\right\rangle}{(aHf(\Omega_m))^2} = -\frac{1}{(aHf(\Omega_m))^2}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial R_2}{\partial r}\right)\right]. \tag{120}$$

The density correlation can be modeled as (with R_2 in Eq. (112))

$$\xi(r,a) = \frac{a_0u^2/(rr_2)}{(aHf(\Omega_m))^2}\cdot\exp\left(-\frac{r}{r_2}\right)\left[\left(\frac{r}{r_2}\right)^2-7\left(\frac{r}{r_2}\right)+8\right]. \tag{121}$$

The derivative of density correlation identifies the peak of correlation

$$\frac{d\xi}{dr} = -\frac{a_0u^2/(r^2r_2)}{(aHf(\Omega_m))^2}\cdot\exp\left(-\frac{r}{r_2}\right)\left[\left(\frac{r}{r_2}\right)^3-8\left(\frac{r}{r_2}\right)^2+8\left(\frac{r}{r_2}\right)+8\right]. \tag{122}$$

Two peaks of correlations can be found at $2r_2$ and $(3+\sqrt{13})r_2 \approx 6.6r_2$, respectively. The mth order moment of the density correlations reads

$$\int_0^\infty \xi(r)\,r^m dr = \frac{a_0u^2r_2^{m-1}}{(aHf(\Omega_m))^2}(m-4)(m-2)\Gamma(m). \tag{123}$$

The averaged correlation $\bar{\xi}(r,a)$ on large scale should read

$$\bar{\xi}(r,a) = \frac{3}{r^3}\int_0^r \xi(y,a)\,y^2dy = -\frac{3}{(aHf(\Omega_m))^2 r}\frac{\partial R_2}{\partial r} = \frac{a_0u^2}{(aHf(\Omega_m))^2}\frac{3}{rr_2}\exp\left(-\frac{r}{r_2}\right)\left(4-\frac{r}{r_2}\right). \tag{124}$$

Equation (124) can be combined with the pair conservation equation [44] to derive the mean pairwise velocity on large scales [18], that is, $\langle\Delta u_L\rangle \propto -Har\bar{\xi}(r,a)$.

Figure 22 presents the comparison of density correlation model in Eq. (121) with $\xi(r)$ from simulation at z=0. The model of $\xi(r)$ derived from velocity correlation function agrees well with simulation on large scale. The model agrees with $\ln[1+\xi(r)]$ even better in a wider range of scale r.

Finally, the correlation of the potential field can be obtained from Eq. (119). For matter dominant model, $H^2 = 8\pi G\rho/3$, the gradient of potential has a correlation

$$R_{\nabla\phi} = \frac{1}{2}\left\langle\nabla\phi(\mathrm{x})\cdot\nabla\phi\left(\mathrm{x}'\right)\right\rangle = \frac{9}{4}\left(\frac{aH}{f(\Omega_m)}\right)^2 R(r) = \frac{9}{8}\left(\frac{aH}{f(\Omega_m)}\right)^2 a_0u^2\exp\left(-\frac{r}{r_2}\right)\left(3-\frac{r}{r_2}\right). \tag{125}$$

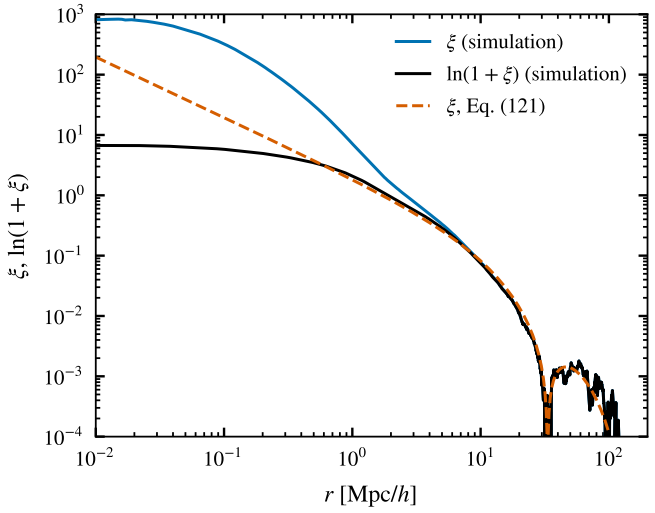


Figure 22. Density correlation at $z = 0$: $\xi(r)$ and $\ln[1+\xi(r)]$ from the simulation, and the large-scale model Eq. (121) derived from the velocity correlations. The model matches ξ on large scales and $\ln(1+\xi)$, the correlation of the log-density field $\eta = \ln(1+\delta)$, over a wider range.

The relations between ϕ and its gradient are (Eq. (93))

$$R_{\nabla\phi} = -\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial R_\phi}{\partial r}\right], \tag{126}$$

$$R_{\nabla^2\phi} = -\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial R_{\nabla\phi}}{\partial r}\right] = \frac{\partial^4 R_\phi}{\partial r^4} + \frac{4}{r}\frac{\partial^3 R_\phi}{\partial r^3}. \tag{127}$$

The correlation function of potential field reads

$$\begin{aligned} R_\phi &= \frac{1}{2}\left\langle \phi(\mathrm{x})\cdot\phi\left(\mathrm{x}'\right)\right\rangle \\ &= \frac{9}{8}\left(\frac{aH}{f(\Omega_m)}\right)^2 a_0 u^2 r_2^2 \exp\left(-\frac{r}{r_2}\right)\left[\left(\frac{r}{r_2}\right)+1\right], \end{aligned} \tag{128}$$

where $R_\phi \propto a^0$ such that the potential fluctuation $\langle\phi^2\rangle$ is independent of time on large scale, i.e., a constant potential perturbation in the linear regime.

5.1.3 *Velocity/density/potential spectrum in Fourier space*

With all the correlation functions developed, the corresponding spectrum functions in the Fourier space can easily be obtained. The velocity spectrum (from Eq. (25)) reads

$$E_u(k) = a_0 u^2 \frac{8}{\pi r_2}\frac{k^{-2}}{\left(1+1/\left(kr_2\right)^2\right)^3}, \tag{129}$$

where $E_u(k) \propto k^4$ for $kr_2 \ll 1$, which is a requirement of the conservation of angular momentum. Note that $E_u(k) \propto k^{-2}$ for $kr_2 \gg 1$ (a signature of Burgers' equation in the weakly nonlinear regime). The maximum $E_u(k)$ is

$$E_u(k_{\max}) = \frac{32}{27\pi} r_2 a_0 u^2 \quad \text{with} \quad k_{\max} r_2 = \sqrt{2}. \tag{130}$$

The spectrum of divergence is obtained from Eq. (118)

$$E_\theta(k) = a_0 u^2 \frac{8}{\pi r_2}\frac{1}{\left(1+1/\left(kr_2\right)^2\right)^3}. \tag{131}$$

The density spectrum $E_\delta(k)$ can be obtained from the density correlation in Eq. (121)

$$\begin{aligned} E_\delta(k) &= \frac{2}{\pi}\int_0^\infty \xi(r,a)\, kr\sin(kr)\, dr \\ &= \frac{16 a_0 u^2}{\left(aHf(\Omega_m)\right)^2 \pi r_2}\frac{1}{\left(1+1/\left(kr_2\right)^2\right)^3}. \end{aligned} \tag{132}$$

The usual matter power spectrum $P_\delta(k,a)$ is related to the density spectrum $E_\delta(k,a)$ as

$$\begin{aligned} P_\delta(k,a) &= 2\pi^2 E_\delta(k,a)/k^2 \\ &= \frac{32\pi a_0 u^2 r_2}{\left(aHf(\Omega_m)\right)^2}\frac{1}{\left(kr_2\right)^2\left(1+1/\left(kr_2\right)^2\right)^3}. \end{aligned} \tag{133}$$

With $k_{\max} r_2 = \sqrt{2}$, the maximum matter power spectrum is

$$P_\delta(k_{\max}, a) = \frac{128\pi a_0 u^2 r_2}{27\left(aHf(\Omega_m)\right)^2}. \tag{134}$$

The spectrum of the potential field $E_\phi(k)$ can be obtained from the Fourier transform of Eq. (128)

$$E_\phi(k) = \frac{18}{\pi r_2}\left(\frac{aH}{f(\Omega_m)}\right)^2 \frac{a_0 u^2 k^{-4}}{\left(1+1/\left(kr_2\right)^2\right)^3}. \tag{135}$$

In this section, the comoving length scale r_2 corresponds to a wavenumber $k_{\max}$ with the maximum power in the density spectrum (see Fig. 23). The pivot wavenumber $k_{\max}$ represents the scale of the particle horizon at the equality epoch [45] that is only dependent on the matter content $\Omega_m h^2$ ($k_{\max} \propto \Omega_m h^2$). Therefore, the comoving scale r_2 should be relevant for the horizon size at the matter-radiation equality. The sound horizon at the drag epoch, i.e., the comoving distance that a sound wave can travel between the big bang and the redshift of the drag epoch, is also related to the wavenumber $k_{\max}$ and depends both on the content of matter and the fraction of the baryon component [45]. Therefore, the length scale r_2 should be related (not equal) to that sound horizon. Figure 23 plots the density spectrum $P_\delta(k)$ that is obtained by the Fourier transform of the density correlation $\xi(r)$ in Fig. 22. The density correlation $\xi(r)$ is obtained directly from the N-body simulation. The corresponding predictions from linear and nonlinear theory are also presented for comparison [31]. Good agreement with the non-linear model in most of the range of k validates the numerical implementation in this work.

5.1.4 *Modeling longitudinal structure function on large scale*

The model for the structure function $S_2^l(r)$ is presented in Eq. (116). With longitudinal correlation function $L_2(r)$ derived on large scales (Eq. (111)), the other second order structure function $S_2^{lp}(r) = \left\langle(\Delta u_L)^2\right\rangle = 2\left(\langle u_L^2\rangle - L_2(r)\right)$ (the dispersion of pairwise velocity Δu_L in Eq. (53)) requires a detailed model of the longitudinal velocity dispersion function $\langle u_L^2\rangle$.

Figure 24 presents the variation of $\langle u_L^2\rangle$ with scale r at different redshift z. Initially $\langle u_L^2\rangle = u^2$ on all scales at high z such that $S_2^{lp}(r) = S_2^l(r)$. With the gradual formation of halo structures, $\langle u_L^2\rangle$ quickly converges to $2u^2$ on small scales where pairs of particles are from the same halo, as discussed in Section 3.3.1 (Eq. (59)). The comoving length scale r_d at which $\langle u_L^2\rangle$ is at its maximum increases with time, where $r_d \approx 0.70 aMpc/h$ (Fig. 11) could be related to the size of the haloes that increases with time.

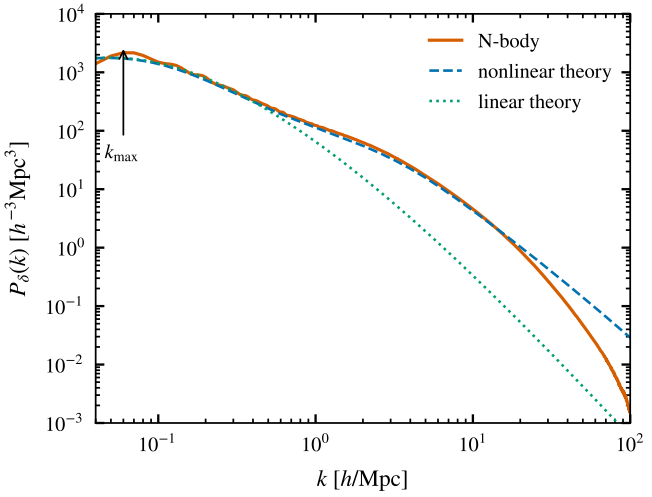


Figure 23. Matter power spectrum $P_\delta(k)$ at $z=0$ from the Fourier transform of the measured $\xi(r)$ (Fig. 22), compared with linear and nonlinear theory [31]. The arrow marks the peak, $k_{\max}$, which corresponds to r_2 ($k_{\max} r_2 = \sqrt{2}$) and depends on $\Omega_m h^2$; the peak value is given by Eq. (134).

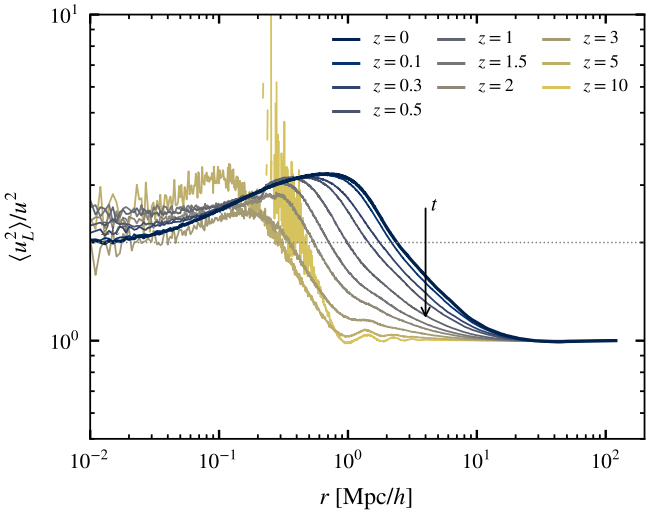


Figure 24. Longitudinal velocity dispersion $\langle u_L^2\rangle$ normalized by $u^2(a)$, for $z=0$–10. At high z, $\langle u_L^2\rangle = u^2$ on all scales and $S_2^{lp} = S_2^l$. As haloes form, $\langle u_L^2\rangle$ approaches $2u^2$ (dotted) on small scales. Its maximum, at $r_d \approx 0.70a\,\mathrm{Mpc}/h$ (Fig. 11), moves to larger scales with time, following the growth of haloes. Model: Eq. (136).

Inspired by the functional form of correlation functions in Section 5.1.1, a good equation to fit $\langle u_L^2\rangle$ on large scale can be found as,

$$\left\langle u_L^2\right\rangle = u^2\left[1 + a_d \exp\left(-\frac{r}{r_{d1}}\right)\left(1-\frac{r}{r_{d2}}\right)\right], \tag{136}$$

where $a_d = 0.44a^{7/4}$, $r_{d1} = 11.953\mathrm{Mpc/h}$ and $r_{d2} = 27.4a^{1/4}\mathrm{Mpc/h}$. The dispersion of pairwise velocity $\left\langle (\Delta u_L)^2\right\rangle$ on large scales can easily be obtained with the Eqs. (136) and (53). Its behavior on small scales follows a two-thirds law that is discussed in a separate paper [41], which can be relevant to dark matter particle mass and properties. Figure 25 presents the variation of $\langle u_L^2\rangle/u^2 - 1$ with scale r at different redshift z, compared with model in Eq. (136).

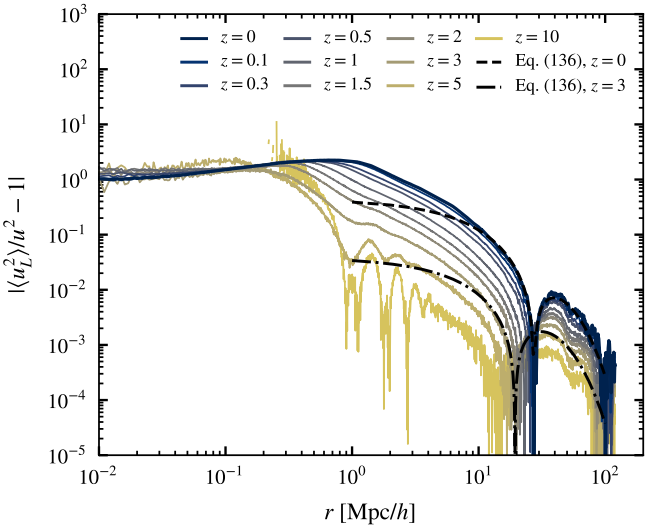


Figure 25. $|\langle u_L^2\rangle/u^2 - 1|$ for $z=0$–10, with the large-scale model Eq. (136) for $z=0$ (dashed) and $z=3$ (dash-dotted). The model describes the data above about 10 Mpc/h at all redshifts.

5.2 Statistical measures on small scale (nonlinear regime)

5.2.1 Structure functions on small scale (nonlinear regime)

The small-scale structure functions follow a power law scaling, as suggested from the N-body simulation data and the virial theorem discussed in Section 4.4. The longitudinal structure function $S_2^l(r) = 2\left(u^2 - L_2(r)\right)$ (Eq. (55)) follows a power law on small scales as $S_2^l(r) \propto r^{1/4}$. Figure 26 plots the variation of $S_2^l(r)$ with scale r at different redshifts z. A power-law scaling $S_2^l(r) \propto r^{1/4}$ is fully developed on small scales after z=1.0 with more haloes formed in the system. It is worth noting the effect of gravitational softening length l_{soft} (black arrow in figure) on the statistics at small scales. The deviation from the 1/4 scaling law was found at scales around or below l_{soft}. To further confirm the statistics on small scales, results from more recent cosmological simulations with a shorter softening length should be explored.

From data in Fig. 26, we model the longitudinal structure function in a general power-law form as

$$S_2^l = 2u^2\,(r/r_1)^n\,, \tag{137}$$

with $n \approx 1/4$ and $r_1(a) \approx r_1^* a^{-3}$ is a length scale dependent on the scale factor a with $r_1^* \approx 19.4 Mpc/h$. The one-fourth power-law scaling of S_2^l was discussed in Section 4.4 (Eq. (109)) that might be related to the power-law scaling of the density correlation $\xi(r)$ via the virial theorem.

5.2.2 Correlation/structure/dispersion functions on small scale

With a structure function of Eq. (137) and kinematic relation in Eq. (82), the longitudinal velocity correlation $L_2(r)$ has the form of

$$L_2(r) = u^2\left[1 - \left(\frac{r}{r_1}\right)^n\right]. \tag{138}$$

Similarly, the kinematic relationships developed in Section 3 can be used to derive the total velocity correlations (Eq. (39)) and transverse

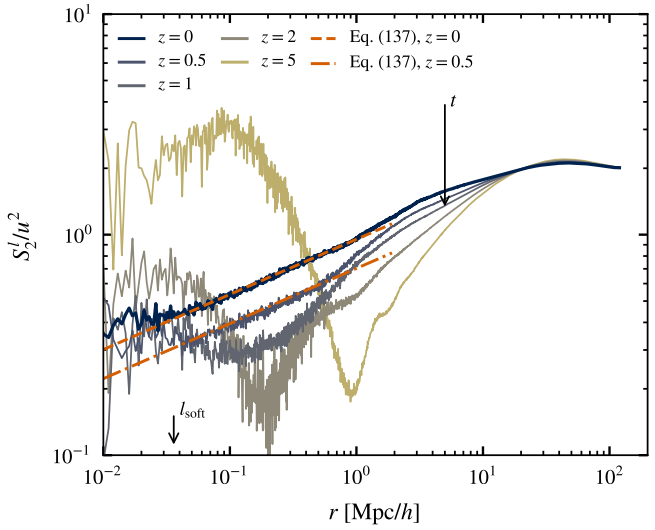


Figure 26. Longitudinal structure function $S_2^l(r)$ normalized by $u^2(a)$ for $z=$ 0, 0.5, 1, 2 and 5. After $z\approx 1$ a one-fourth law, $S_2^l \propto r^{1/4}$, develops on small scales; vermilion lines show Eq. (137) with $n=1/4$ and $r_1 = 19.4a^{-3}$ Mpc/h for $z=0$ (dashed) and $z=0.5$ (dash-dotted). The arrow marks the force-softening length $l_{\rm soft} = 36$ kpc/h (Table 1), below which the measurement is not reliable.

correlation from Eq. (23)

$$R_2 = u^2\left[3-(3+n)\left(\frac{r}{r_1}\right)^n\right], \tag{139}$$

$$T_2 = u^2\left[1-\frac{2+n}{2}\left(\frac{r}{r_1}\right)^n\right]. \tag{140}$$

On small scales, the velocity dispersion function reads (Eq. (30))

$$\sigma_d^2(r) = \frac{24\cdot 2^n}{(4+n)(6+n)}u^2\left(\frac{r}{r_1}\right)^n \approx 1.0745u^2\left(\frac{r}{r_1}\right)^n. \tag{141}$$

The corresponding structure functions are (Eqs. (55) and (67))

$$S_2^l(r) = 2u^2\left(\frac{r}{r_1}\right)^n, \quad S_2^i(r) = 2(3+n)u^2\left(\frac{r}{r_1}\right)^n. \tag{142}$$

The structure function for enstrophy is (from Eq. (76))

$$S_2^x(r) = \frac{6n(3+n)\cdot 2^n}{(4+n)(2+n)}u^2\left(\frac{r}{r_1}\right)^n = 0.6063u^2\left(\frac{r}{r_1}\right)^n. \tag{143}$$

On small scale, the peculiar velocity has a constant divergence such that $R_\theta = 0$. The vorticity correlation can be obtained from Eq. (99)

$$R_\omega = \frac{1}{2}\left\langle \omega(\mathrm{x})\cdot\omega\left(\mathrm{x}'\right)\right\rangle = \frac{n(1+n)(3+n)}{2r^2}u^2\left(\frac{r}{r_1}\right)^n. \tag{144}$$

The corresponding velocity and vorticity spectrum functions are (from Eqs. (25) and (103))

$$E_u(k) = Cu^2 r_1^{-n}k^{-(1+n)}, \quad E_\omega(k) = Cu^2 r_1^{-n}k^{(1-n)}. \tag{145}$$

where with parameters $n=1/4$, $r_1 = 19.4a^{-3}Mpc/h$, the constant

$$C = -\frac{2(3+n)\Gamma((n+3)/2)}{2^{1-n}\Gamma(3/2)\Gamma(-n/2)} = 0.4485. \tag{146}$$

The other two relevant power spectrum for velocity and vorticity fields are $P_u(k)\propto E_u(k)/k^2$ and $P_\omega(k)\propto E_\omega(k)/k^2$.

5.3 Statistical measures on entire range of scales

With correlation functions determined on both small and large scales, we are now ready to model correlation functions for the entire range of flow. The goal is to provide smooth and differentiable velocity correlations for the entire range such that the correlations of vorticity and divergence can be obtained as derivatives of velocity correlations (Eqs. (96) and (97)). The modeling results in this section are good for redshift z=0, while the same method can be extended to model correlation functions at any redshift.

We first introduce a general method to smoothly connect two known functions $f_1(r)$ and $f_2(r)$

$$f_{(fit)}(r) = f_1(r)(1-s(r))^{n1} + f_2(r)s(r)^{n2}, \tag{147}$$

where $s(r)$ is the interpolation function to smoothly connect $f_1(r)$ and $f_2(r)$ with

$$s(r) = \frac{1}{1+x_b e^{-(r-x_c)/x_a}}. \tag{148}$$

With the property that $s(r\gg x_c)=1$ and $s(r\ll x_c)=0$, $f_{(fit)}(r\gg x_c) = f_2(r)$ and $f_{(fit)}(r\ll x_c) = f_1(r)$, where x_a, x_b, x_c, n_1, and n_2 are parameters for each smooth connection.

The velocity correlation function $R_2(r)$ will be first modeled as an example. Here, two functions corresponding to $R_2(r)$ on small and large scales are (Eqs. (139) and (112))

$$f_1(r) = R_{2s}(r) = 3-(3+n)\left(\frac{r}{r_1}\right)^n \tag{149}$$

and

$$f_2(r) = R_{2l}(r) = a_0\exp\left(-\frac{r}{r_2}\right)\left(3-\frac{r}{r_2}\right). \tag{150}$$

The fitted correlation function for the entire range can be expressed as

$$R_{2(fit)}(r) = R_{2s}(1-s(r))^{n1} + R_{2l}(s(r))^{n2}. \tag{151}$$

Here five fitting parameters can be obtained by minimizing the discrepancy between $R_{2(fit)}(r)$ and the correlation function from N-body simulation. The longitudinal and transverse correlations can be modeled in a similar way. The five fitting parameters for three correlation functions are listed in the first three rows of Table 3.

For a better fitting of longitudinal and transverse correlations, an alternative method is proposed. Let us first introduce a new function

$$R_t(r) = 3T_2(r) - R_2(r) = (R_2(r) - 3L_2(r))/2. \tag{152}$$

Once we can fit the function $R_t(r)$ and with fitted $R_2(r)$ from Eq. (151), $L_2(r)$ and $T_2(r)$ can be obtained from Eq. (152)

$$T_2 = (R_t + R_2)/3 \quad \text{and} \quad L_2 = (R_2 - 2R_t)/3. \tag{153}$$

Here we have three segments of function $R_t(r)$

$$R_{ts}(r) = -\frac{n}{2}\left(\frac{r}{r_1}\right)^n, \quad R_{tm}(r) = \frac{r}{r_3} - 0.1, \quad \text{and} \quad R_{tl}(r) = a_0\exp\left(-\frac{r}{r_2}\right)\left(\frac{r}{r_2}\right), \tag{154}$$

where$R_{ts}(r)$ is function on small scale (from Eqs. (138) and (139)) and $R_{tl}(r)$ is a function on large scales (from Eqs. (111) and (112)). The middle segment of the function $R_t(r)$, i.e., $R_{tm}(r)$, is obtained directly from the N-body simulation. The key parameters for correlations in the entire range of r (Eqs. (149), (150) and (154)) are listed in Table 2.

Table 2. Key parameters of the correlation-function models at $z = 0$.

Symbol	Value	Meaning
n	0.25	exponent of S_2^l (Eq. 137)
a_0	0.45	amplitude of T_2 (Eq. 110)
u_0	354.61 km s^{-1}	1D rms velocity (Eq. 1)
r_1	19.4 Mpc/h	scale of S_2^l (Eq. 137)
r_2	23.1321 Mpc/h	scale of T_2 (Eq. 110)
r_3	12.5 Mpc/h	slope of R_{tm} (Eq. 154)
r_t	1.3 Mpc/h	$L_2 = T_2$ crossover

Table 3. Parameters of the smooth connections, Eqs. (147) and (148), used for the all-scale fits at $z = 0$; x_a and x_c in Mpc/h. The fits use $a_0 = 0.4402$ in R_{2l} and R_{tl} (the value in the author's fitting script; Table 2 gives the rounded value 0.45, which changes the fitted curves by at most $0.016u_0^2$). Refitting under the original least-squares objective changes the parameters by less than 0.03 and the residual by less than 0.03 per cent.

Connection	x_a	x_b	x_c	n_1	n_2
$R_{2(fit)}$: $R_{2s} \Leftrightarrow R_{2l}$	2.70	1.35	1.83	0.59	1.09
$L_{2(fit)}$: $L_{2s} \Leftrightarrow L_{2l}$	3.62	2.02	6.36	0.78	3.52
$T_{2(fit)}$: $T_{2s} \Leftrightarrow T_{2l}$	1.28	2.64	0.37	0.30	1.57
R_{t1}: $R_{ts} \Leftrightarrow R_{tm}$	0.21	0.19	0.16	0.5	7.2
R_{t2}: $R_{ts} \Leftrightarrow R_{tl}$	6.05	2.05	11.75	1.39	−0.6
$R_{t(fit)}$: $R_{t1} \Leftrightarrow R_{t2}$	0.5	0.4	3.7	0.9	1.0

A total of three smooth connections must be constructed to fit function $R_t(r)$. First, $R_{t1}(r)$ will be constructed by connecting two functions $R_{ts} \Leftrightarrow R_{tm}$. Second, $R_{t2}(r)$ will be constructed by connecting the small- and large-scale functions $R_{ts} \Leftrightarrow R_{tl}$. Finally, $R_{t(fit)}$ can be constructed by connecting two functions $R_{t1} \Leftrightarrow R_{t2}$. All fitting parameters are also presented in the last three rows in Table 3.

With all fitting parameters listed in Tables 2 and 3, Fig. 27 plots three fitted correlations that agrees very well with the original correlations from N-body simulations. With good models of velocity correlations, the divergence and vorticity correlations and $R_\omega(r)$ can be obtained using Eqs. (96) and (97) in Section 3 with functions $A_2 = (L_2 - T_2)/r^2$ and $B_2 = T_2$ (from Eqs. (16) to (18)). Two correlations of velocity gradients are presented in Fig. 28, where divergence θ is negatively correlated on scale $r > 33$ Mpc/h (Eq. (118)), while vorticity ω is negatively correlated for scale r between 1 Mpc/h and 7 Mpc/h (pair of particles from different haloes) and is positively correlated on small scales $r < 1Mpc/h$.

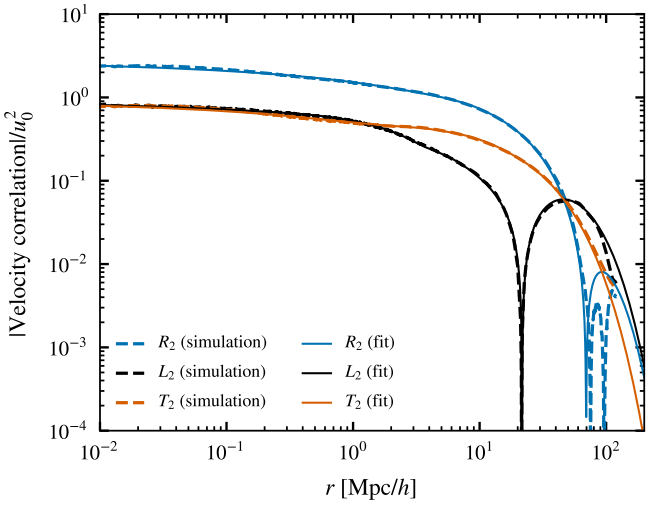


Figure 27. All-scale fits (solid; Eqs. 147–154, parameters in Tables 2 and 3) compared with the measured correlations (dashed) at $z = 0$, normalized by u_0^2; magnitudes are plotted, so sign changes appear as dips.

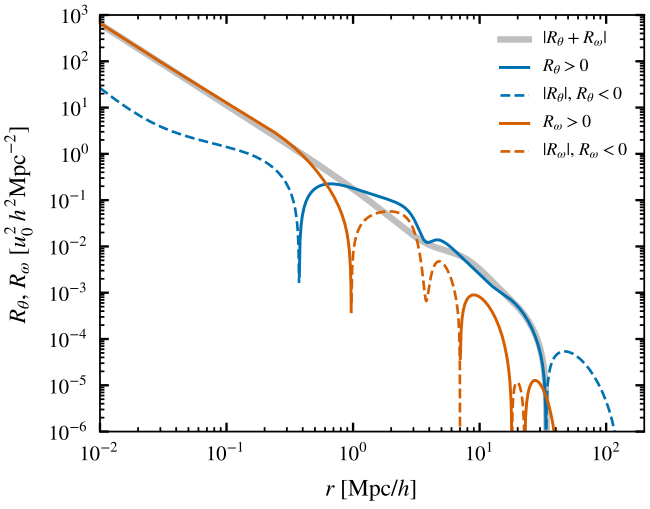


Figure 28. Divergence correlation R_θ and vorticity correlation R_ω at $z = 0$, in units of $u_0^2 h^2$ Mpc^{-2}, computed from the all-scale fits (Eqs. 96, 97). Solid: positive; dashed: magnitude where negative; grey: $|R_\theta + R_\omega|$. R_ω dominates on small scales (constant-divergence flow) and R_θ on large scales (irrotational flow). Vorticity is positively correlated below 1 Mpc/h (pairs in the same halo spin together) and negatively correlated between 1 and 7 Mpc/h; divergence is negatively correlated beyond ≈ 33 Mpc/h.

6 CONCLUSIONS

Treating the velocity field of dark matter as a self-gravitating collisionless fluid, we derived the kinematic relations between second-order statistical measures for three idealized flows and measured those statistics directly from all particle pairs of the Virgo SCDM simulation between $z = 10$ and $z = 0$. The main results are as follows.

(i) *Kinematic relations.* For homogeneous, isotropic flow the correlation, structure, dispersion and spectrum functions are linked by exact relations (Section 3). Incompressible and constant-divergence flows share the same relations for even-order correlations (e.g. Eq. 39) but differ at odd order: $Q_i(r) \equiv 0$ for incompressible flow (Eq. 6), whereas $Q_i(r)$ is constant for constant-divergence flow (Eq. 9). Irrotational flow obeys different relations (Eq. 47).

(ii) *Scale-dependent flow type.* Comparing the volume-averaged correlation R_{2i} with L_2 and T_2 (Fig. 6) shows that the peculiar velocity is a constant-divergence flow on small scales, where pairs share a halo, and irrotational on large scales; the transition is at $r_t \approx 1.3$ Mpc/h at $z = 0$.

(iii) *Energy and enstrophy across scales.* The dispersion functions σ_u^2 and σ_d^2 (Fig. 10) and the real-space distributions E_{ur} and E_{nr} (Figs. 12, 14) show kinetic energy and enstrophy moving to smaller scales as haloes form; half of the kinetic energy lies below $r_{t2} \approx 1$ Mpc/h today.

(iv) *Collisionless signature.* The longitudinal velocities of two particles at vanishing separation are only partly correlated; a halo-

based estimate gives $\rho_L(0) = 1/2$ (Eq. 58), compared with $\rho_L(0) = 1$ for incompressible flow, so the two common definitions of the longitudinal structure function differ ($S_2^{lp} \neq S_2^l$). The smallest separations resolved by the simulation give $\rho_L \approx 0.39$ (Fig. 17). If gravity is the only interaction, a collisionless "annihilation" at $r = 0$ would convert kinetic energy into mass (Eq. 65).

(v) *Large scales.* The transverse correlation is exponential, $T_2 = a_0 u^2 e^{-r/r_2}$ (Eq. 110), with $a_0 u^2 = 0.45 a\, u_0^2$ and a time-independent comoving scale $r_2 \approx 23.1\,\mathrm{Mpc}/h$ that may be related to the horizon size at matter–radiation equality. All other second-order velocity, density and potential correlations and spectra on large scales then follow analytically (Eqs. 111–135); the density model agrees with the measured $\xi(r)$ (Fig. 22).

(vi) *Small scales.* The longitudinal structure function follows a one-fourth law, $S_2^l = 2u^2 (r/r_1)^{1/4}$ with $r_1 = 19.4 a^{-3}\,\mathrm{Mpc}/h$ (Eq. 137), consistent with the virial theorem applied to a power-law density correlation (Eqs. 108, 109). The constant-divergence relations then give the remaining small-scale statistics (Eqs. 138–144), including $E_u(k) \propto k^{-5/4}$ and $E_\omega(k) \propto k^{3/4}$.

(vii) *Velocity gradients.* Smooth all-scale fits (Fig. 27) give the divergence and vorticity correlations (Fig. 28): vorticity is positively correlated below $1\,\mathrm{Mpc}/h$ and negatively correlated between 1 and $7\,\mathrm{Mpc}/h$; divergence is negatively correlated beyond $\approx 33\,\mathrm{Mpc}/h$, which gives the negative density correlation on those scales.

These results rest on a single, relatively low-resolution simulation of one cosmology (SCDM, $\Omega_m = 1$); statistics below the force-softening length ($36\,\mathrm{kpc}/h$) are not reliable. Repeating the analysis on higher-resolution ΛCDM simulations is the natural next test. The kinematics developed here relate statistical measures of the same order; dynamical equations for the velocity field are needed to relate measures of different orders, which is the subject of a companion paper [29].

ACKNOWLEDGEMENTS

This research was supported by the Laboratory Directed Research and Development at Pacific Northwest National Laboratory (PNNL). PNNL is a multiprogram national laboratory operated for the U.S. Department of Energy (DOE) by Battelle Memorial Institute under contract no. DE-AC05-76RL01830.

DATA AVAILABILITY

Two datasets underlying this article, that is, halo-based and correlation-based statistics of dark matter flow, are available on Zenodo [34, 35], along with the accompanying presentation slides 'A comparative study of dark matter flow & hydrodynamic turbulence and its applications' [36]. All data files are also available on GitHub [37]. The correlation-based statistics are Zenodo record 10.5281/zenodo.6757654 (Part II) and the halo-based statistics are record 10.5281/zenodo.6541231 (Part I). The manuscript and the CSV data files behind the figures and numbers of this paper are archived on Zenodo [46].

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